Random Forests for Nonsmooth Binary Choice Models*

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Abstract

We propose a semiparametric estimation procedure for binary choice models using random forests without imposing smoothness or a parametric functional form on the systematic component. Our method uses existing random forest implementations to obtain the tree partitions without additional numerical optimization at each split. The proposed estimator then transforms the adjusted within-leaf sample proportions through the inverse of the known error distribution function and averages the resulting tree-level estimates of the systematic component. We use cross-validation to jointly selects the standard random forest tuning parameters and the additional tuning parameter introduced by our procedure. Under suitable regularity conditions, the theoretical analysis establishes L^2 convergence rates while accommodating continuous and discrete covariates whose dimensions may grow polynomially with the sample size. The simulation results support adapting random forests to the binary choice structure. Compared with the benchmark methods, our proposed estimator performs particularly well when the systematic component exhibits sharp nonlinearities or discontinuities. An empirical application to P2P lending market data demonstrates how the estimator can capture heterogeneous and nonlinear effects on the willingness to default without imposing prespecified credit score categories.

JEL classification: C14, C25.

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1 Introduction

Many economic decisions involve choosing among distinct alternatives, such as products, transportation modes, or education options. Discrete choice models connect these decisions to comparisons of latent utility, allowing researchers to study how preferences respond to changes in economic variables (McFadden, 1974). The random utility framework gives these models a structural microeconomic interpretation by viewing observed choices as outcomes of utility maximization. Binary choice models address the common case of two alternatives, including whether to enter the labor force, enroll in college, adopt a new technology, or participate in a training program (Train et al., 2003; Bhattacharya, 2021). In a standard binary choice model, the utility difference between the two alternatives consists of a systematic component g determined by observed covariates and an unobserved idiosyncratic error component U. An individual chooses an alternative when its utility exceeds that of the other. The distribution F of the idiosyncratic error then serves as a link function connecting the systematic component to the conditional choice probability p.

Classical probit and logit models typically specify g as a linear function of covariates. This specification is parsimonious and easy to interpret, but it may suffer from misspecification when nonlinearities such as interactions and threshold effects are present. These limitations motivate nonparametric approaches that allow g to depart from a predetermined functional form. Sieve methods, such as polynomial or spline approximations, offer one approach. However, capturing complex interactions can require a number of basis functions that grows rapidly with the dimension of the covariates, increasing computational costs and estimation variance. In modern data environments, where researchers often observe many covariates and complex patterns of heterogeneity, machine learning methods provide a natural set of tools for this purpose.

Standard machine learning methods for binary outcomes, however, typically estimate p directly, whereas in a binary choice model g is often the primary object of interest in its own right. The applied economics literature provides strong motivation for discrete choice models and illustrates why g can be the main economic object of interest in many applications. For example, in transportation models, researchers are often interested in measuring willingness to pay for attributes such as shorter travel time, lower tolls, or reduced congestion (Daly et al., 2023). Similarly, in industrial organization, researchers typically construct demand models to study how consumer or product characteristics are translated into monetary values (Gandhi and Nevo, 2021). In such applications, accurate estimation of g is central because it is the object through which economically interpretable welfare comparisons are constructed. p is useful for describing observed choice behavior or market share, but it does not by itself retain the same direct welfare interpretation.

To estimate g with a known F, one valid approach is to estimate p using a standard

machine learning method and subsequently recover g through the plug-in transformation $F^{-1} \circ \widehat{p}$. Such an approach, however, does not exploit the structural information during estimation and thus may entail a cost in finite-sample performance. [1] Consequently, for binary choice models where g is the primary object of interest, it is natural to adapt standard machine learning methods so that the structural information in p is incorporated directly into estimation.

Among many choices of machine learning tools, random forests appear to be particularly attractive. Introduced by Breiman (2001), random forests are tree-based ensemble learning methods that have become some of the most widely used tools for nonparametric regression and classification. Their popularity stems from their ease of tuning, strong predictive performance, computational scalability, and flexibility in capturing complex functional forms such as nonlinearities, discontinuities, and higher-order interactions (Genuer et al., 2008; Lechner and Okasa, 2019). Random forests have become important competitors to other widely used machine learning methods, including neural networks, boosted trees (Freund and Schapire, 1997; Friedman, 2001; Chen and Guestrin, 2016), and support vector machines (Cortes and Vapnik, 1995).

This paper thus studies how random forests can be modified to estimate the binary choice models. We propose an adapted random forest estimator that is simple, computationally efficient, and compatible with existing packages. It begins by treating the binary outcome y_i as a numerical response and growing regression trees using the standard squared-loss CART-split criterion. For each tree, the systematic component within a leaf is estimated by applying the inverse distribution function F^{-1} to the adjusted sample proportion of observations with $y_i = 1$. Because pure leaves generate sample proportions equal to zero or one, for which the F^{-1} transformation is not finite, we introduce two alternative treatments for pure leaves: a clipping method and a pseudo-count adjustment. We consider cross-validation for tuning-parameter

[1] To illustrate, consider squared-loss estimation. A standard machine learning method may solve

$$\widehat{p} = \arg\inf_{p \in \mathcal{P}} \left\{ \frac{1}{n} \sum_{i=1}^{n} (y_i - p(x_i))^2 \right\},$$

where (y_i, x_i) is the training data and $\mathcal{P}$ is a class of probability functions that depends on the specific machine learning algorithm. Of course, direct estimation of p is valid even when the data arise from a binary choice model. However, it does not exploit information about the structural form of p and instead relies on machine learning tools to estimate p nonparametrically.

By comparison, a binary choice model would instead solve

$$\widehat{g} = \arg\inf_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^{n} (y_i - F(g(x_i)))^2 \right\}$$

and then construct $\widehat{p} = F \circ \widehat{g}$. This formulation imposes the known outer transformation F and leaves only g to be estimated flexibly. When the F is correctly or nearly correctly specified, exploiting this additional structure may improve finite-sample estimation of both g and p.

selection tailored to the proposed estimator, which differs from the standard tuning procedure used in random forests.

The paper contributes to the growing literature that uses machine learning methods for economic and econometric analysis. For example, Chernozhukov et al. (2018) develop double/debiased machine learning methods for treatment and structural parameters, using orthogonal scores and sample splitting to mitigate regularization bias from high-dimensional nuisance estimation. Wager and Athey (2018) develop causal forests for estimating heterogeneous treatment effects and provide asymptotic inference theory. Athey et al. (2019) introduce generalized random forests, extending the random forest framework to parameters characterized by local moment conditions. Farrell et al. (2021) study the theoretical properties of deep neural networks as first-stage estimators in semiparametric and causal inference problems. Iskhakov et al. (2020) surveys how machine learning methods, including reinforcement learning, can complement dynamic programming and other tools in structural econometric analysis. More broadly, Varian (2014), Mullainathan and Spiess (2017), and Athey and Imbens (2019) discuss how machine learning methods can complement traditional econometric modeling by improving predictive performance and accommodating complex heterogeneity and high-dimensional covariates. This paper contributes to this literature by adapting random forests to estimation of a structural binary choice model rather than treating the binary outcome solely as a prediction or classification problem.

The paper is also related to the econometric literature on semiparametric and nonparametric estimation of binary choice models. Classical binary response models impose parametric restrictions on g and F. Semiparametric approaches relax some of these restrictions. For instance, Manski (1975) and Horowitz (1992) propose maximum score and smoothed maximum score estimation for discrete choice models, while Klein and Spady (1993) and Ichimura (1993) develop an efficient semiparametric estimator for single-index models. These methods leave F unspecified while retaining a linear index for g. Matzkin (1992) further relaxes the parametric specification of g and studies identification and estimation of nonparametric binary choice models. (Horowitz and Mammen, 2004) proposes an estimator for nonparametric additive g with a known F. These methods, however, are computationally challenging as the sample size increase, and quickly become intractable as number of covariates increases. More recently, Yan (2026) proposes a kernel-based nonparametric estimator for binary choice models using reproducing kernel Hilbert space methods. This approach improves computational scalability in the number of covariates and relaxes parametric restrictions on both g and F, subject to additional identifying conditions.[2] Its estimation theory nevertheless imposes smoothness conditions on the systematic component that exclude discontinuities in g.

[2] Yan (2026) requires g to be a continuous function of the remaining covariates plus one covariate entering linearly with coefficient one.

Our adapted random forest estimator provides a complementary route to flexible binary choice estimation. Because regression trees approximate functions through recursive partitions, they naturally generate piecewise-constant approximations and can accommodate discontinuities, regime changes, threshold effects, and other forms of nonsmooth heterogeneity. Random feature selection and data-adaptive splitting also allow the forest to search over potentially relevant predictors and interactions without requiring the researcher to specify them in advance. These properties make random forests particularly appealing when the systematic component is complex, nonsmooth, or depends on only a subset of a potentially large collection of covariates.

The main contributions of this paper are methodological and theoretical. First, we develop the adapted random forest estimator for semiparametric binary choice models that is computationally effective and easy to tune. Second, we establish convergence properties for the estimators under a set of regularity conditions that allows substantially more flexibility than is typically imposed in the existing random forest literature. The framework permits nonsmooth systematic components and accommodates both continuous and discrete covariates, whose dimensions are allow to grow in polynomial rates. These results provide a theoretical foundation for using random forests for estimation of binary choice models.

Monte Carlo simulations illustrate the circumstances under which our approach is valuable. The adapted random forest estimator performs particularly well when the systematic component contains sharp nonlinearities or threshold discontinuities, while conventional linear models retain their expected advantage when the linear-index specification is correctly specified. The simulations also demonstrate a universal dominance of the proposed estimators over the standard random forest algorithm that estimates the conditional choice probability directly. An empirical application to the P2P loan market also demonstrates the potential of our methodology in capturing the unknown heterogeneous threshold effects.

The rest of the paper is organized as follows. Section 2 reviews Breiman's original random forest algorithm. Section 3 presents the adapted random forest estimator and Section 4 establishes its theoretical guarantees. Section 5 describes the Monte Carlo design and reports the simulation results. Section 6 presents the empirical application and finally, Section 7 concludes.

2 Random forests

Building on the Classification and Regression Trees (CART) framework of Breiman et al. (1984) and the idea of bootstrap aggregation developed by Breiman (1996), random forests combine many randomized, typically unpruned decision trees (Amit and Geman, 1997; Ho, 1998) and aggregate their predictions to reduce variance and improve out-of-sample performance (Bühlmann and Yu, 2002; Biau, 2012; Biau and

Scornet, 2016). During recursive partitioning, each split is selected from a random subset of covariates, which helps decorrelate the individual trees (Hastie et al., 2009). Averaging their predictions can then mitigate overfitting relative to a single decision tree.

In this section, we introduce the concepts and notation needed to describe random forests for both regression and classification. Let $\mathcal{D} = (Y_i, X_i)$ denote n i.i.d. draws from the population of (Y, X), and let $\mathcal{D}_n = (y_i, x_i)$ denote the observed dataset, that is, a realization of $\mathcal{D}$. Let d be the number of covariates.

A *cell* t is defined as a d-dimensional *rectangle* given by

$$t = t_1 \times \cdots \times t_d, \tag{1}$$

where "$\times$" denotes the Cartesian product, and t_j denotes the j-th interval of t. Each t_j belongs to the class of intervals

$$\mathcal{I} = \{[a, b) : 0 \le a \le b \le 1\} \cup \{[a, 1] : 0 \le a \le 1\},$$

that is, the collection of all left-closed, right-open intervals in $[0, 1]$, together with closed intervals in $[0, 1]$ whose right endpoint is one.[3]

A pair $s = (j, c)$, consisting of a covariate index $j \in \{1, \cdots, d\}$ and a cut point $c \in t_j$, is called a *split*. Given a parent cell t, the split $s = (j, c)$ partitions t into two child cells,

$$\begin{aligned} t'(j, c) &= t_1 \times \cdots \times ([0, c) \cap t_j) \times \cdots \times t_d \\ t''(j, c) &= t_1 \times \cdots \times (t_j \cap [c, 1]) \times \cdots \times t_d. \end{aligned} \tag{2}$$

When no confusion arises, we suppress the dependence of the child cells on the split $s = (j, c)$.

A *tree* $T = (t)$ is a collection of pairwise disjoint cells whose union is the unit hypercube $[0, 1]^d$.[4] Finally, given a cell t, define

$$n(t) = \sum_{i=1}^{n} 1\{X_i \in t\} \quad \text{and} \quad \overline{Y}(t) = \frac{1}{n(t)} \sum_{i=1}^{n} 1\{X_i \in t\} Y_i. \tag{3}$$

Here, $n(t)$ is the number of observations falling in t, and $\overline{Y}(t)$ is the sample average of the Y_i's restricted to t. By convention, we set $\overline{Y}(t) = 0$ whenever $n(t) = 0$.

2.1 Random forests for regression

Consider the general nonparametric regression model

$$Y = m(X) + \varepsilon, \tag{4}$$

[3] By definition, $\emptyset \in \mathcal{I}$.

[4] In the consistency analysis, we allow some cells in T to be empty.

where Y is a scalar-valued dependent variable taking values in $\mathbb{R}$, $X = (X_1, \ldots, X_d)$ is a d-dimensional random vector taking values in $[0,1]^d$, $m : [0,1]^d \to \mathbb{R}$ is a measurable nonparametric regression function, and ε is an error term satisfying $\mathbb{E}(\varepsilon|X) = 0$.[5]

We now describe the random forest algorithm. We begin with $[0,1]^d$, referred to as the *root* cell. The root cell is itself a rectangle and is assigned to *level* zero. For a cell t at level ℓ, we randomly select a subset $\theta \subset \{1, \ldots, d\}$ of cardinality $|\theta|$ and split t using the sample CART splitting criterion:

$$s_n = (j_n, c_n) = \underset{\substack{j \in \theta \\ c \in \{x_{ij} : x_i \in t\}}}{\arg\min} \left\{ \sum_{i=1}^{n} \left[1\{x_i \in t'\} \left(y_i - \bar{y}(t')\right)^2 + 1\{x_i \in t''\} \left(y_i - \bar{y}(t'')\right)^2 \right] \right\}, \tag{5}$$

where t' and t'' are defined as in (2).[6] The algorithm breaks ties randomly. The criterion minimizes the within-cell sum of squares across the two child cells.

The optimal split s_n divides t into the child cells t' and t'' defined by (2), which are placed at level $\ell + 1$. After applying this procedure to each cell at level ℓ, the algorithm proceeds to split the cells at level $\ell + 1$. This process continues until all remaining cells satisfy a predetermined stopping rule. A cell that is no longer split is called a *leaf* or terminal cell. This tree-growing procedure is commonly referred to as recursive partitioning, and the random selection of covariates at each split is called column subsampling.

The tree $\widehat{T}(\Theta)$ grown in this way consists of all leaves and forms a partition of $[0,1]^d$. Here, Θ denotes the random variable recording the history of all column subsampling events from the root to the leaves.[7] The *depth* of a tree, denoted by k, is the highest level among its leaves. Throughout the paper, we set $|\theta| = \lceil \gamma d \rceil$ for some $0 < \gamma \leq 1$.[8] Figure 1 illustrates an example of a tree with $T = \{t_{2,1}, \ldots, t_{2,4}\}$ and depth $k = 2$. A tree of depth k has at most 2^k leaves.

[5]The assumption that X takes values in $[0,1]^d$ is without loss of generality and is imposed only for convenience in the theoretical analysis. Normalization of the covariates is not required when implementing the random forest algorithm.

[6]Here, $\bar{y}(t')$ and $\bar{y}(t'')$ minimize the sum of squared errors under a model with a constant prediction in each child cell.

[7]We suppress the dependence of trees on Θ for now and will discuss its structure in detail in the theoretical analysis.

[8]Selecting approximately one-third of the covariates at each split is a conventional choice in random forest regression, although defaults vary across implementations.

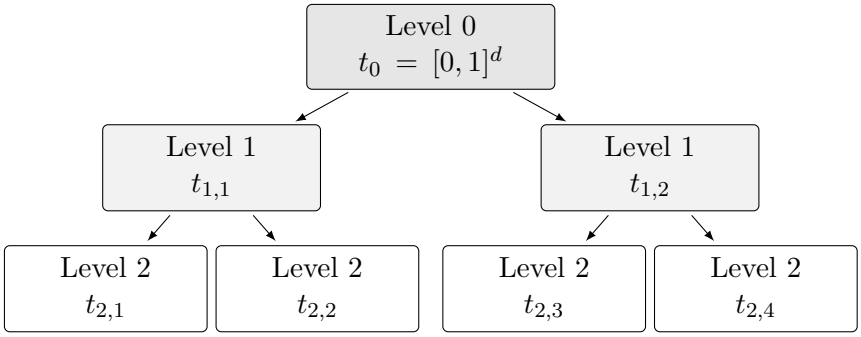


Figure 1: A tree of depth $k = 2$.

Given the training sample $\mathcal{D}_n$, the tree estimator of the regression function m is a simple function over the partition $\widehat{T}$, given by

$$\widehat{m}_{\widehat{T}}(x) = \sum_{t \in \widehat{T}} 1\{x \in t\} \bar{y}(t), \tag{6}$$

for a test point $x \in [0,1]^d$.

The forest estimator $\widehat{m}$ is obtained by averaging A tree estimators:

$$\widehat{m} = \frac{1}{A} \sum_{a=1}^{A} \widehat{m}_{\widehat{T}_a}, \tag{7}$$

where $\widehat{m}_{\widehat{T}_a}$ is the tree estimator constructed from the ath subsample or bootstrap sample of the original training set $\mathcal{D}_n$.

2.2 Random forests for classification

Tree-based methods are also widely used for classification problems. In this paper, we focus on binary classification.

Let $Y \in \{0,1\}$ be a binary random variable. The objects of interest in binary classification are a classifier $K : [0,1]^d \to \{0,1\}$ that predicts the class label and the conditional class probability

$$p(X) = \mathbb{P}(Y = 1|X).$$

Classification trees are grown in the same recursive manner as regression trees, but with a different CART-split criterion. In particular, instead of (5), the optimal split is obtained from

$$s_n = (j_n, c_n) \in \underset{\substack{j \in \theta \\ c \in \{x_{ij} : x_i \in t\}}}{\arg\min} \left\{ n(t') i(\bar{y}(t')) + n(t'') i(\bar{y}(t'')) \right\}, \tag{8}$$

where $i : [0,1] \to \mathbb{R}$ is an impurity measure that is continuous on $[0,1]$ and twice continuously differentiable on $(0,1)$, satisfying (i) $i(0) = i(1) = 0$; (ii) $i(p) = i(1-p)$ for $p \in [0,1]$; and (iii) $i''(p) < 0$ for $p \in (0,1)$.[9] Ties are broken randomly. Two widely used impurity measures are the Gini index, written here up to a positive scaling constant as $i(p) = p(1-p)$, and entropy, $i(p) = -p\log p - (1-p)\log(1-p)$.[10]

For a tree $\widehat{T}$ grown by recursive partitioning using (8), the tree classifier and class-probability estimate at a test point $x \in [0,1]^d$ are, respectively,

$$\widehat{K}_{\widehat{T}}(x) = \sum_{t \in \widehat{T}} 1\{x \in t\} 1\left\{\bar{y}(t) \geq \frac{1}{2}\right\}, \tag{9}$$

$$\widehat{p}_{\widehat{T}}(x) = \sum_{t \in \widehat{T}} 1\{x \in t\} \bar{y}(t). \tag{10}$$

The corresponding forest classifier and class-probability estimator are

$$\widehat{K}(x) = 1\left\{\frac{1}{A}\sum_{a=1}^{A} \widehat{K}_{\widehat{T}_a}(x) \geq \frac{1}{2}\right\}, \tag{11}$$

$$\widetilde{p}(x) = \frac{1}{A}\sum_{a=1}^{A} \widehat{p}_{\widehat{T}_a}(x), \tag{12}$$

where $\widehat{T}_a$ is the tree constructed using the ath subsample or bootstrap sample. Each tree's leaf proportions are computed using its own training sample. Here, we use $\widetilde{p}$ to denote the reduced-form estimator of the conditional choice probability. The estimator $\widehat{p}$ will be defined in Section 3.2. The classifier in (11) uses majority voting, with ties assigned to class one.[11]

2.3 Binary classification as a regression problem

Binary classification can also be viewed as a regression problem:

$$Y = p(X) + \varepsilon, \tag{13}$$

where $Y \in \{0,1\}$, $X = (X_1, ..., X_d) \in [0,1]^d$, and $p : [0,1]^d \to [0,1]$ is a nonparametric function representing the conditional probability that $Y = 1$ given X. Conditional on X, the error term $\varepsilon = Y - p(X)$ has the two-point distribution

$$\varepsilon|_X = \begin{cases} 1 - p(X) & \text{with probability } p(X) \\ -p(X) & \text{with probability } 1 - p(X). \end{cases}$$

[9] See, for example, Section 4.2, particularly Proposition 4.4, of Breiman et al. (1984).

[10] We adopt the convention $0 \log 0 = 0$.

[11] Alternatively, the forest classifier may be defined by thresholding the estimated class probability, i.e., $\widehat{K}(x) = 1\left\{\widetilde{p}(x) \geq \frac{1}{2}\right\}$.

It follows immediately that $\mathbb{E}(\varepsilon|X) = 0$. Thus, (13) is a valid nonparametric regression representation of the binary classification problem.

Because $Y|_X$ follows a Bernoulli distribution, an alternative to the squared-error criterion in (5) is the Bernoulli negative-log-likelihood criterion evaluated at the fitted within-cell probabilities:

$$s_n \in \underset{\substack{j\in\theta \\ c\in\{x_{ij}:x_i\in t\}}}{\arg\min} \left\{ -\sum_{i:x_i\in t'} \left[y_i \log \bar{y}(t') + (1-y_i)\log\{1-\bar{y}(t')\}\right] - \sum_{i:x_i\in t''} \left[y_i \log \bar{y}(t'') + (1-y_i)\log\{1-\bar{y}(t'')\}\right] \right\}. \quad (14)$$

[12]

For binary outcomes, the sample CART-split criterion in (5) is equivalent to the classification split criterion in (8) when the Gini index is used as the impurity measure. To see this, observe that, for any cell $t \subset [0,1]^d$,

$$\begin{aligned}\sum_{i=1}^{n} 1\{x_i \in t\} \left(y_i - \bar{y}(t)\right)^2 &= \sum_{i=1}^{n} 1\{x_i \in t\} y_i^2 - n(t)\bar{y}(t)^2 \\ &= n(t)\bar{y}(t) - n(t)\bar{y}(t)^2 = n(t)\bar{y}(t)\left(1-\bar{y}(t)\right),\end{aligned}$$

where the second equality uses $y_i^2 = y_i$ for $y_i \in \{0,1\}$.[13] Therefore, when $i(p) = p(1-p)$,

$$\begin{aligned}&\underset{\substack{j\in\theta \\ c\in\{x_{ij}:x_i\in t\}}}{\arg\min} \left\{n(t')i(\bar{y}(t')) + n(t'')i(\bar{y}(t''))\right\} \\ &= \underset{\substack{j\in\theta \\ c\in\{x_{ij}:x_i\in t\}}}{\arg\min} \left\{\sum_{i=1}^{n} 1\{x_i \in t'\}\left(y_i - \bar{y}(t')\right)^2 + \sum_{i=1}^{n} 1\{x_i \in t''\}\left(y_i - \bar{y}(t'')\right)^2\right\}.\end{aligned}$$

Analogously, for $i(p) = -p\log p - (1-p)\log(1-p)$,

$$\begin{aligned}&-\sum_{i:x_i\in t} \left[y_i \log \bar{y}(t) + (1-y_i)\log(1-\bar{y}(t))\right] \\ &\quad = -n(t)\left[\bar{y}(t)\log\bar{y}(t) + (1-\bar{y}(t))\log(1-\bar{y}(t))\right].\end{aligned}$$

Hence, (14) is equivalent to (8) with the cross-entropy impurity measure.

Thus, for binary outcomes, least-squares (resp. negative log-likelihood) splitting and Gini-index (resp. cross-entropy) splitting generate the same partitions $\widehat{T}$, up

[12] Here, $\bar{y}(t')$ and $\bar{y}(t'')$ maximize the Bernoulli likelihood under a model with a constant probability in each child cell. Logarithmic terms at the boundaries are interpreted by continuous extension, using $0\log 0 = 0$.

[13] Under the earlier convention that $\bar{y}(t) = 0$ when $n(t) = 0$, this identity also holds for cells containing no training observations.

to random tie-breaking. This equivalence is specific to the binary case and does not extend directly to multiclass classification. Consequently, a binary classification problem can be estimated equivalently by random forest regression with the criterion given by either (5) or (14), and the resulting tree-level and forest-level conditional class probability estimators are obtained as in (10) and (12). The corresponding classifiers are then computed by thresholding these estimated probabilities, as in (9) and (11).

3 Methedology

3.1 The Model

We now describe the binary choice model that generates the binary outcome $Y \in \{0, 1\}$. The model is given by

$$\begin{aligned} Y^* &= g(X) - U \\ Y &= 1\{Y^* > 0\} \end{aligned} \tag{15}$$

where Y^* represents latent utility, $X = (X_1, ..., X_d)$ is a d-dimensional random vector taking values in $[0, 1]^d$, $g : [0, 1]^d \to \mathbb{R}$ is the systematic component belonging to some function class $\mathcal{G}$, and U denotes the idiosyncratic error. We assume that U is independent of X. For a given covariate value $x \in [0, 1]^d$, the conditional choice probability is

$$p(x) = \mathbb{P}(Y = 1|X = x) = \mathbb{E}(Y|X = x) = F(g(x)), \tag{16}$$

where F is the distribution function of U, which is assumed to be known after location and scale normalizations for identification. Typical choices of F include the standard normal distribution in the probit model and the logistic distribution in the logit model.

The objects of interest are the systematic component g and the conditional choice probability $p = F \circ g$. When F is known and strictly increasing, identification of (g, p) reduces to the identification of g. Let P_X denote the probability measure induced by X. We say that $g \in \mathcal{G}$ is identified in $\mathcal{G}$ if, for any $g' \in \mathcal{G}$ such that $F(g(x)) = F(g'(x))$ P_X-a.s., it follows that $g(x) = g'(x)$ P_X-a.s. Thus, with a known strictly increasing F, the systematic component is identified. By contrast, if F is known only up to location and scale, as in a normal distribution with unknown mean and variance, then g is identified only up to the corresponding affine normalization. This is the usual location-scale identification issue in binary choice models.

3.2 Estimation

For the binary choice model in (15), the systematic component g is often estimated by minimizing an empirical loss:

$$\widehat{g} = \underset{g_n \in \mathcal{G}_n}{\arg\inf} \left\{ \sum_{i=1}^{n} \ell(z_i, g_n) \right\}, \tag{17}$$

where $z_i = (y_i, x_i')'$, $\mathcal{G}_n$ is an approximating class contained in $\mathcal{G}$, and ℓ is the loss function. Two natural choices are the squared loss

$$\ell(z, g) = (y - F(g(x)))^2$$

and the negative log-likelihood

$$\ell(z, g) = -y \log F(g(x)) - (1 - y) \log \left(1 - F(g(x))\right).$$

As discussed in Section 2.3, random forests can estimate p directly by treating the binary outcome as a regression target. g can then be estimated via $F^{-1} \circ \widehat{p}$. However, this approach ignores the structural information in p and may come at a cost in finite-sample prediction accuracy, as will be shown in Section 5. We propose a simple and computationally efficient estimator for binary choice models using random forests. The main idea is to approximate g by simple functions defined over tree partitions. Given a partition T of $[0, 1]^d$, a tree-level approximation to g takes the form

$$g_T(X) = \sum_{t \in T} 1\{X \in t\} \beta_t,$$

where β_t is the coefficient associated with t. Similarly, for a parent cell t and a candidate split $s = (j, c)$ that generates child cells t' and t'', the systematic component restricted to t further decomposes as

$$1\{X \in t\} g_T(X) = 1\{X \in t'\} \beta' + 1\{X \in t''\} \beta''.$$

Using the structural relationship $p = F \circ g$, the usual CART-split criterion can be replaced by

$$s_n = \underset{j \in \Theta,\, c \in \{x_{ij} : x_i \in t\}}{\arg\min} \inf_{\beta', \beta'' \in \mathbb{R}} \left\{ \sum_{i : x_i \in t'} \left[y_i - F(\beta') \right]^2 + \sum_{i : x_i \in t''} \left[y_i - F(\beta'') \right]^2 \right\}, \tag{18}$$

for the squared loss in (5) and by

$$\underset{j \in \Theta,\, c \in \{x_{ij} : x_i \in t\}}{\arg\min} \inf_{\beta', \beta'' \in \mathbb{R}} \left\{ - \sum_{i : x_i \in t'} \left[y_i \log F(\beta') + (1 - y_i) \log\{1 - F(\beta')\} \right] \right.$$
$$\left. - \sum_{i : x_i \in t''} \left[y_i \log F(\beta'') + (1 - y_i) \log\{1 - F(\beta'')\} \right] \right\}. \tag{19}$$

for the negative log-likelihood in (14).

At first glance, the criteria in (18) and (19) appear more complicated than the standard CART-split criteria because they involve both the discrete optimization over the candidate split (j, c) and the continuous optimization over the child-node coefficients (β', β''). Since random forests require evaluating a large number of candidate splits across many trees, introducing an additional continuous optimization at every split would substantially increase the computational burden. A key observation is that this complication disappears once the child-node coefficients are profiled out. In particular, the first-order conditions with respect to β' and β'' are

$$F(\beta') = \bar{y}(t') \quad \text{and} \quad F(\beta'') = \bar{y}(t''). \tag{20}$$

Substituting the solutions back into the objective functions, (18) and (19) yield exactly the same optimal splits as (5) and (14), respectively. Consequently, the proposed binary-choice forest can be grown using standard random-forest splitting algorithms without any additional numerical optimization at the splitting stage.

This equivalence is not a generic property of all machine learning methods. Rather, it relies on the cellwise-constant structure of tree estimators and the separability of the split criterion across child cells. The distribution function F affects the transformation from p to g, but it does not alter the optimal split. For other machine learning methods, such as support vector machines or neural networks, incorporating the binary choice structure typically changes the loss function or the optimization problem and thus require new implementations. Random forests therefore have an important computational advantage in estimating binary choice models.

The implementation of the binary-choice forests reduces to two steps. First, run standard random forest regression with binary outcomes using either (5) or (14) and obtain the partitions $(\widehat{T}_a)$ for the entire forest. Second, by the first-order conditions in (20), for the a-th tree $\widehat{T}_a$ constructed from the a-th subsample $\mathcal{D}_a$, the tree estimator of g at a test point x is

$$\widehat{g}_{\widehat{T}_a}(x) = \sum_{t \in \widehat{T}_a} 1\{x \in t\} F^{-1}(\bar{y}(t)). \tag{21}$$

In our implementation, we stop splitting a cell when it has reached level k or when any further split would produce an empty child cell.[14] Then, analogously to (7), the forest estimator is obtained by averaging the tree estimators:

$$\widehat{g}(x) = \frac{1}{A} \sum_{a=1}^{A} \widehat{g}_{\widehat{T}_a}(x). \tag{22}$$

[14]This can occur, for example, when all x_i's in a cell t, restricted to the covariates in θ, are identical. In this case, any candidate split yields no effective variance reduction.

Finally, the plug-in estimator of the conditional choice probability is defined as

$$\widehat{p}(x) = F(\widehat{g}(x)), \tag{23}$$

for $x \in [0,1]^d$. Importantly, notice that (23) is structurally different from the reduced-form estimator given in (12), because in general,

$$F\left(\frac{1}{A}\sum_{a=1}^{A}\sum_{t\in\widehat{T}_a} 1\{x \in t\}F^{-1}(\bar{y}(t))\right) \neq \frac{1}{A}\sum_{a=1}^{A}\sum_{t\in\widehat{T}_a} 1\{x \in t\}\bar{y}(t).$$

In Section 4, we will show that both $\widehat{p}$ and $\widetilde{p}$ are consistent for p. However, as will be shown in Section 5, $\widehat{p}$ has significantly better finite-sample performance than $\widetilde{p}$.

Of course, the definition in (21) requires the within-leaf choice probability $\bar{y}(t)$ to lie strictly between zero and one. As shown in Section 4, under suitable regularity conditions, if the true conditional choice probability is bounded away from zero and one, the probability that $\bar{y}(t)$ falls outside an interval $[\underline{C}, \overline{C}]$ such that $0 < \underline{C} < \overline{C} < 1$ is asymptotically negligible. In finite samples, however, a terminal leaf may be pure, containing only observations with $y_i = 0$ or only observations with $y_i = 1$. In this case, $F^{-1}(\bar{y}(t))$ is undefined. To avoid this numerical problem, we consider two schemes for constructing an interior probability estimate $\widetilde{y}(t)$.

The first approach is a clipping rule. For constants $0 < \underline{C} < \overline{C} < 1$, define

$$\widetilde{y}(t) = \begin{cases} \underline{C} & \text{if } \bar{y}(t) < \underline{C} \\ \bar{y}(t) & \text{if } \underline{C} \leq \bar{y}(t) \leq \overline{C} \\ \overline{C} & \text{if } \bar{y}(t) > \overline{C} \end{cases}. \tag{24}$$

Thus, probabilities below $\underline{C}$ are replaced by $\underline{C}$, while probabilities above $\overline{C}$ are replaced by $\overline{C}$.

Alternatively, we propose a pseudo-count adjustment that modifies only pure terminal leaves. Let

$$\widetilde{y}(t) = \begin{cases} \frac{\sum_{i=1}^{n} 1\{x_i \in t\}y_i + \varphi}{n(t)+2\varphi} & \text{if } \bar{y}(t) \in \{0,1\} \\ \bar{y}(t) & \text{if } 0 < \bar{y}(t) < 1 \end{cases}, \tag{25}$$

where $\varphi > 0$ is a prespecified tuning parameter. With either clipping or pseudo-count adjustment, the estimated probability remains in the interior of the unit interval, so the inverse transformation F^{-1} is well defined.

Implementation The resulting algorithm for the proposed random forest binary choice estimator is straightforward:

1. Fit a standard random forest regression to the binary outcome and obtain the partition generated by each tree grown from the bootstrap sample $\mathcal{D}_a$.

2. For each leaf t of each tree, compute the within-leaf sample proportion $\bar{y}(t)$ and construct the adjusted probability $\tilde{y}(t)$ using either the clipping rule or the pseudo-count adjustment.

3. Transform the adjusted within-leaf probabilities through F^{-1} and construct the tree-level and forest-level estimators using (21), (22), and (23), with $\bar{y}(t)$ replaced by $\tilde{y}(t)$.

Remark 1. For a continuously distributed covariate X_j, if both p and g are differentiable with respect to X_j, the average partial effect (APE) is given by

$$\mathrm{APE}_j = \mathbb{E}\left[\frac{\partial p(X)}{\partial X_j}\right] = \mathbb{E}\left[F'(g(X))\frac{\partial g(X)}{\partial X_j}\right].$$

For a smooth estimator $\widehat{p} = F \circ \widehat{g}$, a natural plug-in estimator is

$$\widehat{\mathrm{APE}}_j = \frac{1}{n}\sum_{i=1}^{n} F'(\widehat{g}(X_i))\frac{\partial \widehat{g}(X)}{\partial X_j}\bigg|_{X=X_i}.$$

The proposed estimator $\widehat{g}$, however, is piecewise constant. Its partial derivative is therefore zero almost everywhere and undefined at the boundaries of the induced partition cells. More importantly, our maintained assumptions do not require either g or p to be differentiable. Consequently, derivative-based marginal effects are not generally well defined within our framework, and directly differentiating the fitted forest yields a degenerate plug-in estimator.

Alternative tools for interpreting the fitted model include partial dependence (PD) functions (Friedman, 2001), individual conditional expectation (ICE) curves (Goldstein et al., 2015), expected forward marginal effects (FME) (Scholbeck et al., 2024), and accumulated local effects (ALE) (Apley and Zhu, 2020).[15] These quantities provide non-derivative interpretations of how a covariate is associated with either the systematic component or the conditional choice probability and are therefore well suited to the nonsmooth structure of tree-based estimators. See, for example, Molnar (2020) for a comprehensive review of machine-learning model interpretation methods.

3.3 Tuning-parameter Selection

An important component of machine learning estimation is the selection of hyperparameters that govern model complexity and predictive performance. In random

[15]For a scalar covariate X_j, the corresponding population functionals may be written as $\mathrm{PD}_j(z) = \mathbb{E}_{X_{-j}}[g(z, X_{-j})]$, $\mathrm{ICE}_{j,x_{-j}}(z) = g(z, x_{-j})$, and $\mathrm{FME}_j(h) = \mathbb{E}[g(X + he_j) - g(X)]$ for a fixed step h and j-th unit vector e_j. In the smooth case, the first-order ALE function is $\mathrm{ALE}_j(z) = \int_{z_0}^{z} \mathbb{E}[\partial g(t, X_{-j})/\partial t | X_j = t]\, dt - C_j$, where C_j is a centering constant. ALE also admits a conditional finite-difference construction that remains applicable to nonsmooth prediction functions. Analogous definitions can be applied to the conditional choice probability $p = F \circ g$.

forests, commonly tuned hyperparameters include the number of candidate covariates considered at each split, the maximum tree depth, and the minimum node size. In the proposed estimator, the clipping parameters or the pseudo-count parameter introduce an additional source of regularization and must therefore be selected jointly with the tree-growing parameters.

A practical feature of random forests is that bootstrap sampling naturally generates out-of-bag (OOB) observations. For each tree, observations that are not selected into its bootstrap sample can be used to evaluate predictive performance, providing an internal validation mechanism without requiring a separate validation sample (Breiman, 2001). The resulting OOB prediction error is therefore commonly used as a computationally inexpensive alternative to K-fold cross-validation (Hastie et al., 2009).

For the estimator proposed in this paper, however, the default OOB criterion reported by standard random forest implementations is not aligned with the prediction error implied by the binary choice model. This distinction is particularly important for the clipping parameters and the pseudo-count parameter. These parameters do not enter the standard OOB error, and thus, the default OOB criterion cannot be used to select them jointly with the tree-growing hyperparameters.

To make this distinction explicit, let ξ denote the collection of tuning parameters, including the tree-growing parameters and the clipping or pseudo-count parameter. Under squared loss, the validation error is

$$\frac{1}{n_{\text{val}}} \sum_{i=1}^{n_{\text{val}}} \Big(y_i - \widehat{p}(x_i;\xi)\Big)^2,$$

where

$$\begin{aligned}\widehat{p}(x_i;\xi) &= F\left(\widehat{g}(x_i;\xi)\right) \\ &= F\left(\frac{1}{A}\sum_{a=1}^{A}\sum_{t\in\widehat{T}_a} 1\{x_i \in t\} F^{-1}\Big(\widetilde{y}(t;\xi)\Big)\right)\end{aligned}$$

with $\widetilde{y}(t;\xi)$ defined in (24) and (25). As discussed in Section 3.2, this criterion is generally different from that associated with $\widetilde{p}$. Though a structurally modified OOB criterion could in principle be constructed by applying the same transformation and aggregation rule to the trees for which each observation is out of bag, such a criterion would need additional theoretical justification. We therefore employ K-fold cross-validation, which directly evaluates the forest estimator for p.

Let $\mathcal{I}_1, \ldots, \mathcal{I}_K$ be a partition of the training-sample indices. The folds are constructed by stratified sampling so that the proportion of observations with $y_i = 1$ is approximately the same in each fold as in the full training sample. For each k, the forest estimator $\widehat{p}^{(-k)}(\cdot;\xi)$ is fitted using the observations whose indices are not in $\mathcal{I}_k$ and is evaluated on the held-out fold $\mathcal{I}_k$.

The K-fold cross-validation error is defined as

$$\mathrm{CV}(\xi) = \frac{1}{n_{\text{train}}} \sum_{k=1}^{K} \sum_{i \in \mathcal{I}_k} \left[y_i - \widehat{p}^{(-k)}(x_i; \xi) \right]^2. \tag{26}$$

Equivalently,

$$\mathrm{CV}(\xi) = \sum_{k=1}^{K} \frac{|\mathcal{I}_k|}{n_{\text{train}}} \mathrm{CV}_k(\xi),$$

where

$$\mathrm{CV}_k(\xi) = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left[y_i - \widehat{p}^{(-k)}(x_i; \xi) \right]^2.$$

The weighting by $|\mathcal{I}_k|/n_{\text{train}}$ accommodates the possibility that stratification produces folds of slightly different sizes. When all folds have equal size, this expression reduces to the unweighted average $K^{-1} \sum_{k=1}^{K} \mathrm{CV}_k(\xi)$.

The selected tuning parameter vector is

$$\arg\min_{\xi \in \Xi} \mathrm{CV}(\xi),$$

where Ξ denotes the prespecified tuning grid. After the optimal ξ is selected, the forest is re-estimated using the full training sample. Tuning-parameter selection based on negative loglikelihood loss can be defined analogously.

4 Theoretical Properties

In this section, we establish L^2 convergence rates for the forest estimators $\widehat{p}(X)$ and $\widehat{g}(X)$. Building on the theoretical frameworks of Chi et al. (2022) and Mazumder and Wang (2023), our analysis extends earlier consistency results along three directions. First, we do not require the conditional choice probability p or the systematic component g to be smooth. Second, we allow X to contain both continuous and discrete components. Third, the numbers of both types of covariates may increase with the sample size at polynomial rates. In contrast, much of the existing theory assumes continuously distributed covariates, imposes smoothness or structural restrictions on the regression function such as Lipschitz continuity or additivity, and treats the dimension d as fixed (Biau, 2012; Scornet et al., 2015; Wager and Athey, 2018). Such assumptions are useful for controlling approximation error, but they can be restrictive in econometric applications characterized by mixed covariate types, increasing dimensionality, and nonsmooth target functions.

To accommodate nonsmoothness, we work with the sufficient-impurity-decrease (SID) function class introduced by Chi et al. (2022). The flexibility for nonsmooth target parameter is relevant in many economic applications, where discontinuities can arise from institutional rules, eligibility requirements, policy thresholds, regime changes, and other forms of discrete decision making. Threshold and regime-switching models, for example, are designed precisely to capture abrupt changes in economic relationships, while regression discontinuity designs exploit discontinuous changes in treatment assignment at known cutoffs (Hansen, 2000; Imbens and Lemieux, 2008; Lee and Lemieux, 2010).

In addition, we extend the results of Chi et al. (2022) and Mazumder and Wang (2023) by allowing the covariate vector to contain both continuous and discrete components, with the dimensions of both components permitted to grow at arbitrary polynomial rates with the sample size. These extensions are particularly relevant for empirical economics, where researchers often work with high-dimensional collections of covariates that combine continuous characteristics with discrete indicators such as gender, race, education categories, and treatment status. A theoretical framework restricted to continuously distributed covariates or fixed-dimensional settings therefore excludes many specifications routinely encountered in applied work. By accommodating mixed covariate types together with increasing dimensionality, our analysis more closely reflects the structure of modern economic data and the environments in which machine learning methods are commonly applied.

4.1 Proof Strategy

We now outline the proof strategy. Following the framework of Chi et al. (2022), we grow each tree on a subsample drawn without replacement. The main object of analysis is the L^2 prediction error of a single-tree estimator of the reduced-form conditional choice probability,

$$\mathbb{E}\left(\widehat{p}_{\widehat{T}}(X) - p(X)\right)^2,$$

where $\widehat{p}_{\widehat{T}}$ is defined in (10).

First, because the forest estimator is obtained by averaging the corresponding tree estimators, the convergence analysis of the forest estimator can be reduced to that of the individual trees. In particular, by the definition of the forest estimator in (22) and Jensen's inequality,

$$\mathbb{E}\left(g(X) - \widehat{g}(X)\right)^2 \leq \frac{1}{A}\sum_{a=1}^{A}\mathbb{E}\left(g(X) - \widehat{g}_{\widehat{T}_a}(X)\right)^2,$$

where $\widehat{g}_{\widehat{T}_a}$, $a = 1, \cdots, A$ denotes the tree estimator constructed from the a-th subsample. Therefore, any upper bound on the L^2 error of the individual tree estimators

immediately yields an upper bound of the same order for the forest estimator.[16]

Let each tree be grown on a subsample of size $n' = \lceil bn \rceil$ with $0 < b \leq 1$. Since n' is proportional to n, the rate obtained for a tree grown from a subsample is of the same order as the rate obtained from the full sample, up to constant multiples depending on b. Hence, it suffices to derive the L^2 convergence rate for a single tree estimator grown from the full training sample $\mathcal{D}_n$. The corresponding results for a subsample-grown tree follow by replacing n with $n' = \lceil bn \rceil$ throughout the argument.

Second, we will show that under suitable regularity assumptions, the convergence of $\widehat{g}_{\widehat{T}}$ defined in (21) with $\bar{y}(t)$ replaced by $\widetilde{y}(t)$ follows from the convergence of $\widehat{p}_{\widehat{T}}$, that is,

$$\mathbb{E}\left(\widehat{g}_{\widehat{T}}(X) - g(X)\right)^2 \leq C\mathbb{E}\left(\widehat{p}_{\widehat{T}}(X) - p(X)\right)^2.$$

Accordingly, the remainder of the analysis focuses on deriving the L^2 convergence rate of the reduced-form tree estimator $\widehat{p}_{\widehat{T}}$ obtained from standard random forest regression.

The fundamental difficulty in the asymptotic analysis of random forests is the data-dependent nature of the partition $\widehat{T}$. In particular, the usual bias-variance decomposition of the mean squared error,

$$\mathbb{E}\left(\mathbb{E}\widehat{p}_{\widehat{T}}(X) - p(X)\right)^2 \quad \text{and} \quad \mathbb{E}\left(\widehat{p}_{\widehat{T}}(X) - \mathbb{E}\widehat{p}_{\widehat{T}}(X)\right)^2,$$

are difficult to analyze because $\widehat{T}$ itself is random and depends on the data. Following the strategy of Chi et al. (2022), we instead introduce a semi-population tree estimator,

$$p^*_{\widehat{T}}(X) = \sum_{t \in \widehat{T}} 1\{X \in t\}\mathbb{E}(Y|X \in t), \tag{27}$$

which keeps the sample-generated partition $\widehat{T}$ fixed but replaces the sample average in each leaf by its population counterpart. This object allows us to decompose the L^2 error into two terms by:

$$\mathbb{E}\left(p^*_{\widehat{T}}(X) - p(X)\right)^2 \quad \text{and} \quad \mathbb{E}\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2. \tag{28}$$

Although (28) is not the classical bias-variance decomposition, it plays an analogous role. We refer to the first term as the approximation error, or bias term, because it measures how well a piecewise-constant function over rectangular-based partition

[16] This argument does not require independence across trees and therefore does not exploit the dependence structure among the tree estimators. A sharper analysis of the cross-tree covariance structure may yield tighter convergence bounds and potentially reveal additional gains from forest averaging. We leave such an analysis for future research.

approximates the conditional expectation of $p(X)$. We refer to the second term as the stochastic error, or variance term, because it measures the error from estimating the within-leaf population means by the corresponding sample analogs.

To analyze these two terms, we first rewrite the sample CART-split criterion in a variance-reduction form. For a parent cell t and a split $s = (j, c)$ that divides t into child cells t' and t'', the squared-loss CART criterion in (5) can be equivalently written as

$$\begin{aligned} s_n &= \underset{j\in\Theta, c\in\{X_{ij}:X_i\in t\}}{\arg\min} \left\{ \sum_{i:X_i\in t'} \left(Y_i - \overline{Y}(t')\right)^2 + \sum_{i:X_i\in t''} \left(Y_i - \overline{Y}(t'')\right)^2 \right\} \\ &= \underset{j\in\Theta, c\in\{X_{ij}:X_i\in t\}}{\arg\max} Q_n(t, s), \end{aligned} \tag{29}$$

where $Q_n(t, s)$ is defined by

$$Q_n(t, s) = \frac{1}{n} \sum_{i:X_i\in t} \left(Y_i - \overline{Y}(t)\right)^2 - \frac{1}{n} \sum_{i:X_i\in t'} \left(Y_i - \overline{Y}(t')\right)^2 - \frac{1}{n} \sum_{i:X_i\in t''} \left(Y_i - \overline{Y}(t'')\right)^2.$$

Thus, $Q_n(t, s)$ measures the sample variance reduction achieved by splitting t into t' and t''.

It can be easily shown that criterion function in (29) is the sample analog of its population version $Q(t, s)$ defined by

$$Q(t, s) = \mathbb{P}\{X \in t\}\text{var}(Y|X \in t) - \mathbb{P}\{X \in t'\}\text{var}(Y|X \in t') - \mathbb{P}\{X \in t''\}\text{var}(Y|X \in t''). \tag{30}$$

The corresponding population CART-split criterion is

$$s^* = \underset{j\in\Theta, c\in t_j}{\arg\max}\ Q(t, s). \tag{31}$$

Intuitively, the population CART split chooses the split that maximizes the population variance reduction within the parent cell.

In the theoretical analysis, however, the representations in (29) and (31) are still not the most convenient to work with directly. Following Chi et al. (2022) and Mazumder and Wang (2023), we use the following equivalent representation.

Lemma 1. *For any cell t and split s of t, we have*

$$\begin{aligned} Q(t, s) = \mathbb{P}\{X \in t'\} \left(\mathbb{E}(Y|X \in t') - \mathbb{E}(Y|X \in t)\right)^2 \\ + \mathbb{P}\{X \in t''\} \left(\mathbb{E}(Y|X \in t'') - \mathbb{E}(Y|X \in t)\right)^2. \end{aligned} \tag{32}$$

Proof. See Appendix B.1 for the proof of Lemma 1. □

Lemma 1 shows that the population CART criterion can be equivalently interpreted as the weighted dispersion of the conditional means in the child cells around the conditional mean in the parent cell. This representation also allows us to interpret the splitting rule as recursively maximizing the between-cell heterogeneity captured by each split.

We now summarize how the two terms in (28) are controlled. For the variance term, we use the identity

$$\mathbb{E}\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2 = \mathbb{E}\left[\sum_{t\in\widehat{T}} \left(\overline{Y}(t) - \mathbb{E}(Y|X\in t)\right)^2 \mathbb{P}\{X\in t\}\right].$$

This term is a sum of local estimation errors over the partition made by rectangles, where the local estimation errors are the differences between the within-leaf sample averages and their population counterparts. The proof constructs a high-probability event E_1 on which the local sample averages are uniformly close to their population counterparts across all relevant cells. On A, the stochastic error is bounded using concentration inequalities. On the complement E_1^c, the error is controlled by showing that $\mathbb{P}(E_1^c)$ is asymptotically negligible.

For the bias term, note that

$$\mathbb{E}\left(p^*_{\widehat{T}}(X) - p(X)\right)^2 = \mathbb{E}\left[\sum_{t\in\widehat{T}} \mathbb{P}\{X\in t\}\,\mathrm{var}\left(p(X)|X\in t\right)\right].$$

This term measures the remaining within-leaf variation of the conditional choice probability after the tree reaches a given depth. To illustrate the idea, temporarily ignoring column subsampling, the proof constructs a high-probability event E_2 on which the sample CART-split criterion is uniformly close to its population counterpart. On this event, the sample CART achieves nearly the same impurity reduction as the population CART. A recursive argument from the leaves at level k back to the root cell $[0,1]^d$ then yields a vanishing upper bound for the remaining within-leaf variation. On the complement E_2^c, the convergence rate is again controlled by the negligibility of $\mathbb{P}(E_2^c)$. The formal proof incorporates column subsampling by accounting for the probability that an informative covariate is included among the candidate split variables at each cell.

Combining the stochastic-error bound and the approximation-error bound gives the L^2 convergence rate for the tree estimator $\widehat{p}_{\widehat{T}}$. By the preceding arguments, this rate then implies the corresponding convergence rates for the forest estimators $\widehat{g}$ and $\widehat{p}$.

Remark 2. In what follows, we clarify the meaning of the objects $\mathbb{E}(Y|X\in t)$ and $\mathbb{P}\{X\in t\}$, which appear frequently in the literature and in this paper but require some

care when the cell t is data-dependent. If t is fixed, there is no ambiguity: $\mathbb{E}(Y|X \in t)$ and $\mathbb{P}\{X \in t\}$ are deterministic quantities. In the CART algorithm, however, t is constructed from the training sample $\mathcal{D}$ and the column-subsampling randomness Θ. Hence, the event $\{X \in t\}$ is generally not $\sigma(X)$-measurable, but is measurable with respect to $\sigma(X, \mathcal{D}, \Theta)$. The population quantities associated with the realized cell should therefore be understood conditionally on $(\mathcal{D}, \Theta)$ and are $\sigma(\mathcal{D}, \Theta)$-measurable random variables.

To formalize this interpretation, recall the following measure-theoretic construction. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For any event $E \in \mathcal{F}$ such that $\mathbb{P}(E) > 0$, define $\mathbb{P}_E(A) = \frac{\mathbb{P}(A \cap E)}{\mathbb{P}(E)}$ for $A \in \mathcal{F}$. Then $\mathbb{P}_E$ is a probability measure on the measurable space $(\Omega, \mathcal{F})$. Moreover, for any $\mathcal{F}$-measurable and $\mathbb{P}_E$-integrable random variable Y, we have

$$\mathbb{E}_{\mathbb{P}_E}(Y) = \int Y d\mathbb{P}_E = \frac{1}{\mathbb{P}(E)} \int Y 1_E d\mathbb{P} = \mathbb{E}(Y|E).$$

More generally, for any σ-field $\mathcal{G} \subset \mathcal{F}$,

$$\mathbb{E}_{\mathbb{P}_E}(Y|\mathcal{G}) := \mathbb{E}(Y|E, \mathcal{G}) = \frac{\mathbb{E}(Y 1_E|\mathcal{G})}{\mathbb{P}(E|\mathcal{G})}$$

$\mathbb{P}_E$-a.s., is a valid conditional expectation, with the ratio defined arbitrarily on the set $\{\omega : \mathbb{P}(E|\mathcal{G}) = 0\}$. To verify the defining property, for every $G \in \mathcal{G}$,

$$\begin{aligned}
\int_G \frac{E(Y 1_E|\mathcal{G})}{\mathbb{P}(E|\mathcal{G})} d\mathbb{P}_E &= \frac{1}{\mathbb{P}(E)} \int \frac{E(Y 1_E|\mathcal{G})}{\mathbb{P}(E|\mathcal{G})} 1_E 1_G d\mathbb{P} \\
&= \frac{1}{\mathbb{P}(E)} \mathbb{E}\left(\frac{E(Y 1_E|\mathcal{G})}{\mathbb{P}(E|\mathcal{G})} \mathbb{E}(1_E|\mathcal{G}) 1_G \right) \\
&= \frac{1}{\mathbb{P}(E)} \mathbb{E}\left(\mathbb{E}(Y 1_E 1_G|\mathcal{G}) \right) \\
&= \frac{1}{\mathbb{P}(E)} \int (Y 1_G) 1_E d\mathbb{P} = \int_G Y d\mathbb{P}_E.
\end{aligned}$$

Accordingly, when t is generated from $(\mathcal{D}, \Theta)$, throughout the theoretical analysis, we interpret

$$\mathbb{P}\{X \in t\} := \mathbb{P}(X \in t|\mathcal{D}, \Theta) \quad \text{and}$$

$$\mathbb{E}(Y|X \in t) := \begin{cases} \mathbb{E}(Y|X \in t, \mathcal{D}, \Theta) = \dfrac{\mathbb{E}(Y 1\{X \in t\}|\mathcal{D}, \Theta)}{\mathbb{P}(X \in t|\mathcal{D}, \Theta)}, & \text{if } \mathbb{P}(X \in t|\mathcal{D}, \Theta) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Conditional on a realization of $(\mathcal{D}, \Theta)$, the cell t is fixed, and these quantities reduce to the usual population probability and conditional expectation associated with that realized cell.

For simplicity, we suppress the conditioning on $(\mathcal{D}, \Theta)$ whenever a data-dependent cell t is involved and set $\mathbb{E}(Y|X \in t) = 0$ whenever $\mathbb{P}\{X \in t\} = 0$.

4.2 Assumptions

We are now ready to state the assumptions used in the theoretical analysis.

Assumption 1. *There exists some* $\lambda \in (0, 1]$ *such that for each cell* $t \subset [0, 1]^d$,

$$\sup_{j \in \{1, \cdots, d\}, c \in t_j} Q(t, s) \geq \lambda \mathbb{P}\{X \in t\} \operatorname{var}\left(p(X) \middle| X \in t\right) \tag{33}$$

Assumption 1 is the core condition on the conditional choice probability p and the distribution of the covariates X. It is used in Chi et al. (2022) and Mazumder and Wang (2023), where it is referred to as the sufficient impurity decrease (SID) condition. The left-hand side of (33) is the unrestricted population CART split criterion, in the sense that all covariates are considered as candidate splitting variables. The SID condition states that, for every cell t, the best population variance reduction achieved by a single split is at least a fixed positive fraction of the original within-cell variation of $p(X)$. Thus, the condition guarantees that p can be well approxiamted through recursive partitioning.

Importantly, the SID condition does not require p to be continuous, differentiable, or otherwise smooth. The parameter λ measures how effectively CART can approximate the target function through axis-aligned binary splits. A larger value of λ implies that each split removes a larger fraction of the remaining within-cell variation. Intuitively, functions that are close to simple functions, or whose main variation can be captured by a small number of axis-aligned splits, should satisfy Assumption 1 with a relatively large value of λ. For example, if $p(X)$ is a one-dimensional single-threshold step function, then it is easy to see that $\lambda = 1$.

For more general functions, however, the SID condition can be difficult to verify, and deriving an explicit value of λ may be technically involved.[17][18] Nevertheless, SID provides a useful formalization of the requirement that CART splits be sufficiently informative. We leave further relaxation of this condition for future research.

Assumption 2. $\mathcal{D}_n = (Y_i, X_i)$ *consists of i.i.d. copies of* (Y, X).

[17] Examples of functions satisfying SID are provided in Chi et al. (2022), although the corresponding derivations can be nontrivial. Mazumder and Wang (2023) consider additive functions of the form $p(X) = \sum_{j=1}^{d} p_j(X_j)$ and provide more tractable sufficient conditions for SID through a locally reverse Poincaré-type inequality.

[18] Another limitation of Assumption 1 is that, for many function classes, the value of λ may deteriorate as the number of informative covariates increases. Hence, the convergence rate may become slower in complex high-dimensional settings.

Assumption 2 is imposed to simplify the concentration arguments used in the proof. In particular, Lemma 4 relies on concentration inequalities such as Hoeffding's inequality and Bernstein's inequality to control the deviations between sample quantities and their population counterparts uniformly over a large collection of cells and candidate splits. Although the present paper focuses on the i.i.d. setting, the proof strategy suggests that an extension to weakly dependent data may be possible by replacing these concentration inequalities with suitable dependent-data analogues. Such an extension would be useful for panel data and time-series applications, but we leave it for future research.

Assumption 3. *Denote by d_1 and d_2 the numbers of continuous-type and discrete-type covariates, respectively.*

(i) $d_1 = O\left(n^{C_1}\right)$ *and* $d_2 = O\left(n^{C_2}\right)$ *for some positive constants* C_1 *and* C_2.

(ii) The cardinality of the support of every discrete-type covariate is finite.

Assumption 3 differs from much of the existing random forest theory, which is often developed for continuous covariates only.[19] By contrast, Assumption 3(i) allows the numbers of both continuous-type and discrete-type covariates to grow at arbitrary polynomial rates, which is a more practical feature in modern empirical applications. Assumption 3(ii) is imposed mainly for technical simplicity. It can be relaxed to allow the support size of each discrete covariate to grow with the sample size, provided that the growth rate is sufficiently controlled.

Assumption 4. *The joint probability density function for the continuous covariates, denoted by f, is bounded away from* 0 *and* ∞.

Assumption 4 is a standard regularity condition in nonparametric regression with continuous covariates. It prevents the distribution of the continuous covariates from being too sparse or too concentrated in particular regions of the covariate space and allows the probability masses of rectangular cells to be controlled in our theoretical analysis. Assumption 4 is weaker than requiring the continuous covariates to be jointly uniformly distributed on $[0,1]^{d_1}$, as is often assumed in the random forest literature (Biau, 2012; Wager and Athey, 2018).

Assumption 5. *The conditional choice probability is uniformly bounded away from zero and one, i.e.,* $\underline{C} \leq p(X) \leq \overline{C}$ *X-a.s. for some constants* $\underline{C}$ *and* $\overline{C}$ *such that* $0 < \underline{C} \leq \overline{C} < 1$.

[19] For example, Biau (2012) and Scornet et al. (2015) study consistency in a fixed-dimensional continuous-covariate settings, while Chi et al. (2022) and Mazumder and Wang (2023) allow high-dimensional covariates but maintain a continuous-covariate framework.

Assumption 6. *The distribution function F of the idiosyncratic error U is strictly increasing and admits a density that is bounded away from zero on every compact subset of $\mathbb{R}$.*

Assumption 5 is common in discrete choice analysis (Horowitz, 1992; Chang et al., 2026). Together with Assumption 6, it ensures that the systematic component $g = F^{-1} \circ p$ is bounded and allows the convergence rate of $\widehat{p}_{\widehat{T}}$ to be transferred to $\widehat{g}_{\widehat{T}}$ through the Lipschitz continuity of F^{-1} on a compact interval. Strict monotonicity of F is standard in both parametric and nonparametric binary choice models (Matzkin, 1992; Yan, 2026). The probit and logit models satisfy Assumption 6, since their densities are strictly positive and continuous on every compact interval.

Assumption 7. *The tree $\widehat{T}$ satisfies the following regularity condition: there exists $0 < \alpha < \frac{1}{2}$ such that for large enough n, it holds that $\frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\} \geq n^{-\alpha}$ for every $t \in \widehat{T}$.*

Assumption 7 is a technical condition that controls the size of leaves. Equivalently, it requires every leaf to contain at least $n^{1-\alpha}$ observations for all sufficiently large n. Since $0 < \alpha < 1/2$, each leaf contains more than $n^{1/2}$ observations asymptotically.[20] Together with Assumption 5, this condition is used to show that $\overline{Y}(t)$ is uniformly bounded away from zero and one with high probability. This ensures that $F^{-1}(\overline{Y}(t))$ remains well behaved asymptotically.

Regularity conditions controlling leaf sizes and tree geometry are common in the random forest literature. For example, Wager and Athey (2018) and Friedberg et al. (2020) impose related conditions that restrict how unbalanced each split can be. Assumption 7 differs from these conditions in that we do not require every split to allocate a fixed fraction of the observations to each child node; instead, we directly impose a lower bound on the size of every terminal cell. Moreover, Assumption 7 could be further relaxed by requiring the condition to hold only on a high-probability event whose complement is asymptotically negligible at an appropriate rate.

The next section states the main convergence results and discusses the key ideas behind their proofs.

4.3 Main Results

Throughout, we will use C to denote a generic positive constant whose value may change from line to line. For an arbitrary set A, denote its complement by A^c and its cardinality by $|A|$. Finally, for two numerical sequences a_n and b_n, write $a_n \sim b_n$ if $a_n/b_n \to 1$ as $n \to \infty$.

[20] Assumption 7 implicitly restricts the tree depth k from growing too quickly. This is compatible with our theoretical analysis, in which we consider shallow trees and set the tree depth to $k = \lfloor \eta \log_\delta n \rfloor$ for some $\delta > 1$ and $\eta > 0$.

As discussed in Section 4.1, the analysis of the tree estimator is based on the following bias-variance-type decomposition:

$$\mathbb{E}\left(\widehat{p}_{\widehat{T}}(X)-p(X)\right)^2 \leq 2\mathbb{E}\left(p^*_{\widehat{T}}(X)-p(X)\right)^2 + 2\mathbb{E}\left(\widehat{p}_{\widehat{T}}(X)-p^*_{\widehat{T}}(X)\right)^2, \tag{34}$$

where $p^*_{\widehat{T}}$ is defined in (27). We first establish separate convergence rates for these two terms and then combine them to obtain the L^2 convergence rate for $\widehat{p}_{\widehat{T}}$. Finally, we transfer the rate from $\widehat{p}_{\widehat{T}}$ to $\widehat{g}_{\widehat{T}}$ through F^{-1}.

Theorem 1. *Let Assumptions 2-4 hold. Let $0 \leq k \leq \lfloor \eta \log_\delta n \rfloor$ for some $\eta > 0$ and $\delta > 1$. Then, it holds that*

$$\mathbb{E}\left(p^*_{\widehat{T}}(X)-\widehat{p}_{\widehat{T}}(X)\right)^2 = O\Big(2^k n^{-1}(\log n)^3(\log\log n)^2\Big).$$

In addition, if $k = \lfloor \eta \log_\delta n \rfloor$, then

$$\mathbb{E}\left(p^*_{\widehat{T}}(X)-\widehat{p}_{\widehat{T}}(X)\right)^2 = O\Big(n^{-1+\eta\log_\delta 2}(\log n)^3(\log\log n)^2\Big).$$

Proof. See Appendix C.1 for the proof of Theorem 1. □

Theorem 1 controls the stochastic error arising from estimating the within-leaf conditional means by their sample analogues. The proof of Theorem 1 uses a uniform concentration argument over all cells that can be generated by at most k splits. Notice that

$$\mathbb{E}\left(\widehat{p}_{\widehat{T}}(X)-p^*_{\widehat{T}}(X)\right)^2 = \mathbb{E}\left[\sum_{t\in\widehat{T}}\left(\overline{Y}(t)-\mathbb{E}(Y|X\in t)\right)^2\mathbb{P}\{X\in t\}\right].$$

The proof constructs a high-probability event A such that, on A, the weighted local estimation error

$$\left(\overline{Y}(t)-\mathbb{E}(Y|X\in t)\right)^2\mathbb{P}\{X\in t\}$$

is uniformly bounded by $Cn^{-1}(\log n)^3(\log\log n)^2$ over all cells generated by at most k splits. Since $\widehat{T}$ contains at most 2^k leaves, summing over the terminal cells gives the stated rate. On the complement A^c, the local error is bounded by a constant, while $\mathbb{P}(A^c)$ decays faster than any polynomial rate in n. Therefore, the contribution from A^c is asymptotically negligible.

This argument does not depend on the particular CART-split criterion or on column subsampling. It uses only the fact that the estimator is a simple function over a partition consisting of at most 2^k rectangular cells. Consequently, the rate in Theorem 1 is not specific to CART and does not involve the parameters γ or λ. These parameters enter the approximation error rather than the stochastic error.

To ensure that the stochastic error vanishes, the depth k cannot grow too quickly. A logarithmic depth choice, $k = \lfloor \eta \log_\delta n \rfloor$, is natural because it gives $2^k \sim n^{\eta \log_\delta 2}$. The constants η and δ jointly control tree depth. k increases as η increases or δ decreases. For consistency, it is required that $\eta \log_\delta 2 < 1$.

Theorem 1 alone would favor trees with fixed or slowly growing depth. However, trees that are too shallow may not approximate a complex conditional choice probability sufficiently well. The bias term studied below shows that allowing the tree depth to grow at an appropriate rate helps reduce the approximation error. Hence, overall consistency would reflect a trade-off between stochastic error and approximation error through the choice of tree depth.

Theorem 2. *Let Assumption 1-4 hold. Then, given $\gamma, \lambda \in (0, 1]$ and $k = \lfloor \eta \log_\delta n \rfloor$ with $\eta > 0$ and $\delta > 1$, it holds that*

$$\mathbb{E}\left(p^*_{\widehat{T}}(X) - p(X)\right)^2 = \begin{cases} O\left(n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)} + n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2\right) & \text{if } \gamma\lambda \neq 1 \\ O(n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2) & \text{if } \gamma\lambda = 1 \end{cases} \tag{35}$$

Proof. See Appendix C.2 for the proof of Theorem 2. □

Theorem 2 gives the rate for the approximation error. The rate in (35) contains two components. The first component,

$$n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)},$$

is the population approximation term, which corresponds to the idealized recursive population CART procedure defined in (31). Under the SID condition (Assumption 1), each split removes at least a λ fraction of the remaining within-cell variation when all covariates are available. With column subsampling, only a fraction of covariates is considered at each split, and the effective variance reduction is governed by $\gamma\lambda$. Thus, after k levels, the remaining approximation error behaves like $(1-\gamma\lambda)^k$, which decays as k grows. When $k = \lfloor \eta \log_\delta n \rfloor$, this term becomes $n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)}$. This expression has the expected comparative statics. A larger λ means that the population CART-split criterion $Q(t, s)$ is more effective at reducing variance, and hence the approximation error decreases faster. A larger γ means that more covariates are considered at each split, making it more likely that an informative split is available.

The second component,

$$n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2,$$

comes from replacing the population CART split $Q(t, s)$ by the sample CART split $Q_n(t, s)$ at each cell. Since $Q_n(t, s)$ is only a sample analog of $Q(t, s)$, the empirical

split s_n may not coincide exactly with the population split s^*. The proof therefore requires a uniform bound on the discrepancy between the sample and population split criteria over all relevant cells and candidate splits. This error accumulates across cells and leads to the second term. As in the stochastic error bound, the factor $2^k \sim n^{\eta \log_\delta 2}$ appears because the number of cells grows with tree depth.

When $\gamma\lambda = 1$, the optimal population split effectively eliminates the within-cell variation at each level, so the first component disappears from the displayed rate. The remaining approximation error is then driven by the empirical approximation of the population split criterion. In this case, the approximation bound is still slightly larger than the stochastic-error bound in Theorem 1 as it carries an additional logarithmic factor.

Remark 3. The proof of Theorem 2 is based on a recursive variance-reduction argument. For clarity, we first ignore column subsampling and set $\gamma = 1$. The argument extends to the case with column subsampling, which is treated in the proof of Theorem 2 in Appendix C.2. Let $\widehat{T}_\ell$ denote the collection of cells at level ℓ. The approximation error can be written as

$$\mathbb{E}\left(p^*_{\widehat{T}}(X) - p(X)\right)^2 = \mathbb{E}\left[\sum_{t \in \widehat{T}_k} \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t)\right].$$

This identity shows that the approximation error is the expected within-leaf variation of $p(X)$. The goal of recursive partitioning is to make this quantity small.

For a parent cell t at level $k-1$ with child cells t' and t'', we get

$$\sum_{t \in \widehat{T}_k} \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) \tag{36}$$

$$= \sum_{t \in \widehat{T}_{k-1}} \left(\mathbb{P}\{X \in t'\}\mathrm{var}(p(X)|X \in t') + \mathbb{P}\{X \in t''\}\mathrm{var}(p(X)|X \in t'')\right) \tag{37}$$

$$= \sum_{t \in \widehat{T}_{k-1}} \left(\mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) - Q(t, s_n)\right) \tag{38}$$

Equation (37) follows from rearranging the summands in (36), while (38) follows from the definition of the sample split s_n and the equivalent representation of $Q(t, s)$,

$$\begin{aligned} &Q(t,s) \\ &= \mathbb{P}\{X \in t\}\mathrm{var}(Y|X \in t) - \mathbb{P}\{X \in t'\}\mathrm{var}(Y|X \in t') - \mathbb{P}\{X \in t''\}\mathrm{var}(Y|X \in t'') \\ &= \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) - \mathbb{P}\{X \in t'\}\mathrm{var}(p(X)|X \in t') - \mathbb{P}\{X \in t''\}\mathrm{var}(p(X)|X \in t''). \end{aligned}$$

If the sample split s_n were identical to the population split s^*, then Assumption 1 would imply

$$(38) \leq \sum_{t \in \widehat{T}_{k-1}} (1-\lambda)\mathbb{P}\{X \in t\}\mathrm{var}(Y|X \in t) \tag{39}$$

Iterating this inequality from level $k-1$ back to the root yields

$$(39) \leq (1-\lambda)^k \mathrm{var}(p(X)) = O\left((1-\lambda)^k\right).$$

In practice, however, $Q(t, s_n)$ and $Q(t, s^*)$ are not equal. The formal proof addresses this issue by constructing a high-probability event B on which $Q(t, s_n)$ and $Q(t, s^*)$ are uniformly close. More precisely, for any $\epsilon > 0$, the recursive bound takes the form

$$\begin{aligned}(38) &\leq \sum_{t \in \widehat{T}_{k-1}} \left(1 - \frac{\lambda}{(1+\epsilon)^2}\right) \mathbb{P}\{X \in t\} \mathrm{var}(p(X)|X \in t) + C\epsilon^{-1} 2^{k-1} n^{-1} (\log n)^3 (\log\log n)^2 \\ &\leq \left(1 - \frac{\lambda}{(1+\epsilon)^2}\right)^k \mathrm{var}(p(X)) + C\epsilon^{-1} 2^k n^{-1} (\log n)^3 (\log\log n)^2\end{aligned}$$

on event B. Setting $\epsilon = (\log n)^{-1}$ and $k = \lfloor \eta \log_\delta n \rfloor$ yields the rate in Theorem 2.

The following result combines the stochastic- and approximation-error bounds and gives the overall L^2 convergence rate for $\widehat{p}_{\widehat{T}}$, which follows immediately from the triangle inequality given in (34).

Corollary 1. *Let Assumption 1-4 hold. Then, given $\gamma, \lambda \in (0, 1]$, and $k = \lfloor \eta \log_\delta n \rfloor$ with $\eta > 0$ and $\delta > 1$, it holds that*

$$\mathbb{E}\left(p(X) - \widehat{p}_{\widehat{T}}(X)\right)^2 = \begin{cases} O\left(n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)} + n^{-1+\eta \log_\delta 2} (\log n)^4 (\log\log n)^2\right) & \text{if } \gamma\lambda \neq 1 \\ O(n^{-1+\eta \log_\delta 2} (\log n)^4 (\log\log n)^2) & \text{if } \gamma\lambda = 1 \end{cases}. \tag{40}$$

The result shows that the convergence rate for the tree estimator can be slow when γ is close to zero. Column subsampling, however, is commonly used in random forests both to reduce computational cost and to decorrelate individual trees, which may reduce the variance of the forest estimator; see, for example, Hastie et al. (2009, Chapter 15). Our tree-level analysis does not capture this potential benefit from forest averaging. Thus, although column subsampling worsens the convergence bound for an individual tree in our analysis, it may still improve the performance of the overall forest. A sharper characterization of the role of column subsampling, particularly its interaction with dependence across trees, is left for future research.

Next, we state the convergence result for the tree estimator $\widehat{g}_{\widehat{T}}$.

Theorem 3. *Let Assumption 1-7 hold. Then, given $\gamma, \lambda \in (0, 1]$, and $k = \lfloor \eta \log_\delta n \rfloor$ with $\eta > 0$ and $\delta > 1$, it holds that*

$$\mathbb{E}\left(g(X) - \widehat{g}_{\widehat{T}}(X)\right)^2 = \begin{cases} O\left(n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)} + n^{-1+\eta \log_\delta 2} (\log n)^4 (\log\log n)^2\right) & \text{if } \gamma\lambda \neq 1 \\ O(n^{-1+\eta \log_\delta 2} (\log n)^4 (\log\log n)^2) & \text{if } \gamma\lambda = 1 \end{cases}. \tag{41}$$

Proof. See Appendix C.3 for the proof of Theorem 3. □

The key additional step is to transfer the convergence rate from the tree estimator $\widehat{p}_{\widehat{T}}$ to $\widehat{g}_{\widehat{T}} = F^{-1} \circ \widehat{p}_{\widehat{T}}$. Under Assumptions 5–7, with high probability the within-leaf averages remain away from zero and one. In particular, there exists an event E_3 with probability tending to one such that, on E_3,

$$\frac{\underline{C}}{2} \leq \overline{Y}(t) \leq \frac{1+\overline{C}}{2}$$

for every $t \in \widehat{T}$, where $\underline{C}$ and $\overline{C}$ are defined in Assumption 5. Since F^{-1} is Lipschitz continuous on $[\underline{C}/2, (1+\overline{C})/2]$, it follows that

$$\mathbb{E}\left[\left(\widehat{g}_{\widehat{T}}(X) - g(X)\right)^2 1(E_3)\right] \leq C\mathbb{E}\left[\left(\widehat{p}_{\widehat{T}}(X) - p(X)\right)^2 1(E_3)\right]$$

Hence, the estimator $\widehat{g}$ inherits the same convergence rate as the reduced-form tree estimator $\widehat{p}_{\widehat{T}}$.

Finally, we give the convergence result for the forest estimator $\widehat{g}$. In particular, if λ were known, one could choose the tree depth k to balance the two leading powers of n and obtain an optimized convergence rate.

Theorem 4. *Let Assumption 1-4 hold. Then, given $\gamma, \lambda \in (0,1]$, $n' = \lceil bn \rceil$ with $b \in (0,1]$, and $k = \lfloor \eta \log_\delta n \rfloor$ with $\eta > 0$ and $\delta > 1$, it holds that*

$$\mathbb{E}\left(g(X) - \widehat{g}(X)\right)^2 = \begin{cases} O\left(n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)} + n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2\right) & \text{if } \gamma\lambda \neq 1 \\ O(n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2) & \text{if } \gamma\lambda = 1 \end{cases}. \tag{42}$$

If in addition, $\eta = \left(\log_\delta \frac{2}{1-\gamma\lambda}\right)^{-1}$, then it holds that

$$\mathbb{E}\left(g(X) - \widehat{g}(X)\right)^2 = O\left(n^{-1+\frac{1}{1-\log_2(1-\gamma\lambda)}}(\log n)^4(\log\log n)^2\right). \tag{43}$$

Proof. See Appendix C.4 for the proof of Theorem 4. □

By the Lipschitz continuity of F, the convergence rate of $\widehat{p}$ is implied by that of $\widehat{g}$ through $\widehat{p} = F \circ \widehat{g}$.

5 Monte Carlo Simulation

This section evaluates the finite-sample performance of the proposed random forest estimator. We compare our method with three benchmark estimators. The simulation designs are deliberately heterogeneous, and we therefore do not expect any estimator to dominate uniformly across all designs.

5.1 Simulation Design

The data generating process (DGP) considered here is a binary choice model with $U \overset{iid}{\sim} \mathbb{N}(0,1)$. Because the error distribution is set to be standard normal, the proposed estimator is referred to as the Random Forest Probit (RFP) estimator.

We consider four functional specifications for the systematic component g. The second specification is evaluated at two values of its shape parameter, giving five simulation designs in total:

$$
\begin{aligned}
(1): \quad & g(X) = \sum_{j=1}^{5} \beta_j \sin\left(\pi X_j X_{j+5}\right) + \sum_{j=1}^{5} \beta_{j+5} X_j X_{j+5}, \\
(2): \quad & g(X) = \alpha_\kappa \sum_{j=1}^{10} \beta_j \left[\frac{1}{1+\exp(-\kappa X_j)} - \frac{1}{2}\right], \qquad \kappa \in \{1, 100\}, \\
(3): \quad & g(X) = \sum_{j=1}^{10} \beta_j \left[1\{X_j > 0\} - \frac{1}{2}\right], \\
(4): \quad & g(X) = \sum_{j=1}^{10} \beta_j X_j.
\end{aligned}
$$

We let $X_j \overset{iid}{\sim} \text{Unif}[-1,1]$ for all j. For specifications (1)–(3), $X = (X_1, \ldots, X_{10})'$, whereas for specification (4), $X = (X_1, \ldots, X_{20})'$. Thus, specification (4) contains ten relevant and ten irrelevant covariates. The symmetry and centering of the specifications imply that $\mathbb{P}\{Y = 1\} = 1/2$ under every design.

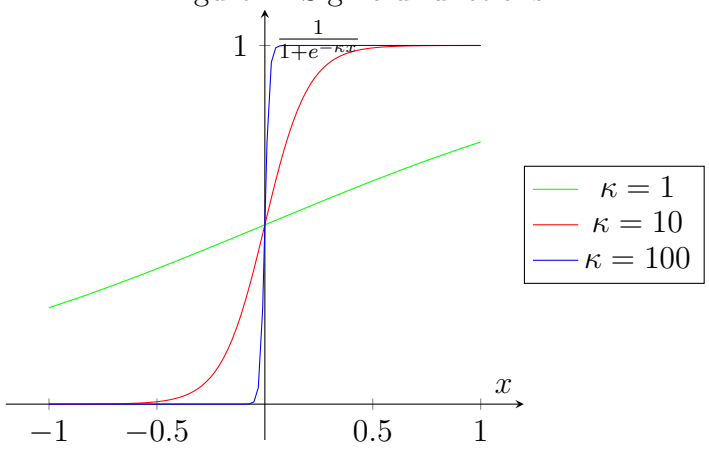
Across all specifications, we use

$$(\beta_1, \ldots, \beta_{10}) = (-0.47, -0.26, 0.15, 0.82, -0.60, 0.80, 0.89, 0.32, 0.26, -0.88).^{21}$$

Specification (1) is smooth but highly nonlinear and contains pairwise interactions. It is designed to examine whether RFP can approximate a smooth interaction function. Specification (2) provides a controlled transition from an approximately linear function to one with very sharp changes. The sigmoid function is close to linear on $[-1,1]$ when $\kappa = 1$. As κ increases, the sigmoid becomes increasingly steep and approaches a threshold-type response. Figure 2 illustrates the shape of the sigmoid function for different values of κ.

[21] The coefficients are generated independently from Unif$[-1,1]$ using seed = 1 in R and are rounded to two decimal places.

Figure 2: Sigmoid functions



To make the sigmoid designs comparable across κ, we normalize the variance of the systematic component to one. In particular, $\text{var}(g(X)) = 1$ when

$$\alpha_\kappa = \left[\frac{1}{4}\left(1 - \frac{2\tanh(\kappa/2)}{\kappa}\right)\sum_{j=1}^{10}\beta_j^2\right]^{-1/2}.$$

Specification (3) is a purely discontinuous design in which every relevant covariate enters through a threshold at zero. Finally, in specification (4), we consider a linear probit model where half of the covariates are irrelevant. This specification is used to assess the robustness of the competing methods to irrelevant covariates.

For RFP, we select the tuning parameters jointly by stratified 5-fold cross-validation. Let $n_{\min}$ be the minimum leave size. In particular, we stop splitting a cell if it contains less than n_{min} number of observations. Denote by m_{try} the size of column subsampling, which is the cardinality of θ defined in Section 2.

The candidate grids are

$$\begin{aligned}
n_{\min} &\in \{5, 25, 50, 75, 100\},\\
m_{\text{try}} &\in \left\{\lfloor\sqrt{d}\rfloor, \lfloor d/3\rfloor, d\right\},\\
k &\in \{\lfloor 0.5\log_2 n\rfloor, \lfloor\log_2 n\rfloor, \lfloor 1.5\log_2 n\rfloor, \lfloor 2\log_2 n\rfloor, \infty\},\\
\varphi &\in \{0.001, 0.1, 1\},
\end{aligned}$$

where k is the maximum tree depth and φ is the pseudo-count parameter defined in (25) in Section 3.2.[22] We fix the number of trees at $A = 500$. Each tree is grown from a bootstrap sample of the full training data. We use bootstrap sampling rather than subsampling so that no additional subsample-fraction tuning parameter is introduced;

[22] Repeated values in the m_{try} grid are removed.

this choice is made for computational convenience. For diagnostic purposes, we also compute the standard random forest regression that estimate p directly as in (12) and obtains $\widehat{g}$ through $F^{-1} \circ \widehat{p}$, with tuning parameters selected from the same grids as those used for RFP without φ.

The first nonparametric benchmark is the Kernelized Probit (KP) estimator, considered as a special case of the kernelized procedure proposed by Yan (2026). The estimator approximates g in a Gaussian reproducing kernel Hilbert space (RKHS) with kernel

$$K(v, w) = \exp\left(-\frac{\|v - w\|^2}{2\sigma^2}\right),$$

where the bandwidth is fixed at $\sigma = 1$. Estimation uses squared probability loss subject to a RKHS norm constraint B_n^2 and spectral-cutoff parameter m for regularization. We select the constraint and cutoff parameters by 5-fold cross-validation over

$$\begin{aligned} B_n^2 &\in \{5, 10, 20, 30, 40, 50, 75, 100, 200, 500\}, \\ m &\in \{10, 25, 50, 75, 100, 150, 200\}.^{23} \end{aligned}$$

We also consider the second-order polynomial probit (SPP) model as well as the linear probit (LP) method. The SPP specifies the approximation

$$g_{\text{SPP}}(x) = \beta_0 + \sum_{j=1}^{d} \beta_j x_j + \sum_{j=1}^{d} \beta_{jj} x_j^2 + \sum_{1 \le j < \ell \le d} \beta_{j\ell} x_j x_\ell.$$

Both SPB and LP are estimated by maximum likelihood with a probit link and an intercept.

For each DGP and Monte Carlo replication, we generate a training sample with $n \in \{1{,}000, 2{,}000, 5{,}000\}$ and a test sample with $n_{\text{test}} = 10{,}000$. We conduct 1,000 Monte Carlo replications.

For evaluation metrics, we consider the root mean squared errors (RMSE) and the mean absolute deviations (MAD) respectively given by

$$\begin{aligned} \text{RMSE}(\widehat{g}) &= \left[\frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left(\widehat{g}(x_i) - g(x_i)\right)^2\right]^{\frac{1}{2}} \quad \text{and} \\ \text{MAD}(\widehat{g}) &= \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left|\widehat{g}(x_i) - g(x_i)\right|. \end{aligned}$$

And we calculate the same quantities for $\widehat{p}$.

[23] These candidate grids follow the implementation considered by Yan (2026).

5.2 Simulation Results

Intuitively, when the DGP is a binary choice model, estimating p directly using a machine learning method remains valid, but ignoring the model's structural information can come at a cost in finite-sample performance. Tables 1 and 2 illustrate this point for random forests. The RFP estimator outperforms standard random forest regression across all DGPs and sample sizes considered, achieving lower RMSE and MAD for both the systematic component g and the choice probability p. These results support the finite-sample benefits of incorporating the binary choice structure into random forest estimation.

Tables 3 and 4 report the simulation results for all probit methods. The RMSE and MAD criteria yield the same broad conclusions. Most importantly, no estimator dominates uniformly across designs. For the sinusoidal specification in Panel A, KP and SPP outperform RFP, while LP performs substantially worse than the other three estimators. At $n = 1{,}000$, KP has the smallest RMSE for $\widehat{g}$, whereas SPP has a slightly smaller MAD for $\widehat{g}$ and smaller RMSE and MAD for $\widehat{p}$. At the larger sample sizes, SPP performs best under all four criteria. This ranking is consistent with the structure of specification (1). The systematic component is smooth and contains pairwise products $X_j X_{j+5}$, which enter directly into the SPP basis. Although the sinusoidal terms are not quadratic, their smooth variation over the compact covariate support is reasonably well captured by the polynomial and kernel approximations. In contrast, RFP approximates the smooth interaction surface using rectangular step functions and may therefore require a relatively fine partition to attain comparable accuracy.

For the sigmoid specification with $\kappa = 1$ in Panel B, LP performs best under every criterion at every sample size. This result is expected because, over $[-1, 1]$, the sigmoid function with $\kappa = 1$ is approximately linear. LP is therefore nearly correctly specified and benefits from estimating relatively few parameters. SPP contains the linear specification as a special case, but estimating its additional quadratic and interaction terms can be costly in small samples. Its performance improves substantially as the sample size increases.

The ranking reverses when $\kappa = 100$. In Panel C, RFP has the smallest RMSE and MAD for both $\widehat{g}$ and $\widehat{p}$ at every sample size, and its relative advantage becomes more pronounced as the sample size increases. The performance of the other three estimators deteriorates substantially as κ increases, and their errors decline more slowly with sample size. This pattern is consistent with approximation or regularization bias remaining important over the sample sizes considered. Sharp nonlinearities that closely resemble discontinuities pose greater challenges for the global polynomial and smooth kernel approximations used here, whereas tree-based methods can accommodate such features through recursive partitioning. Panel D displays a similar pattern under genuine discontinuities: RFP again performs best under every criterion

at every sample size.

Finally, Panel E considers an exactly linear systematic component in the presence of ten irrelevant covariates. As expected, LP performs best by a wide margin because it is correctly specified and parsimonious. Although RFP cannot match LP's accuracy, it outperforms KP and SPP. SPP is also correctly specified in this design, but its performance illustrates the cost of estimating many unnecessary parameters. With $d = 20$, a complete second-order basis contains 231 coefficients, including the intercept, which can lead to substantial estimation variance at smaller sample sizes. For KP, a plausible explanation for its weaker performance is that irrelevant covariates enter the Gaussian kernel's distance calculations despite containing no information about the outcome. RFP can partially mitigate this difficulty by favoring informative covariates in its data-dependent split selection. These results suggest that RFP remains competitive relative to KP and SPP in the presence of irrelevant predictors.

6 Empirical Application

This section illustrates the proposed estimator using loan-default outcomes from the peer-to-peer (P2P) lending market, where both lenders and borrowers can virtually meet for loan transactions. In contrast to traditional banks, which often rely on opaque internal algorithms, P2P platforms operate via a transparent "marketplace" model. In this setting, the platform assigns risk grades and interest rates to borrowers primarily based on hard credit information, most notably the FICO score. Because lending platforms may use discrete score thresholds when assigning credit grades and pricing loans, the relationship between FICO scores and default risk need not be smooth. In particular, the borrower's default disutility may exhibit threshold effects or discontinuities at credit-score values that are relevant to the platform's underwriting algorithm. Detecting such features is therefore important for understanding borrower risk and the decision rules embedded in lending markets.

A common approach in the empirical literature is to model default probabilities using logistic regression and to introduce FICO scores through researcher-specified score categories. For example, Emekter et al. (2015) divide FICO scores into six intervals, 640–659, 660–678, 679–713, 714–749, 750–779, and 780 or above. Similarly, Jagtiani and Lemieux (2019) use five FICO intervals: 650–679, 680–699, 700–749, 750–800, and 800 or above. These specifications impose the locations of the relevant thresholds ex ante, even though the exact score cutoffs used by lending platforms in their underwriting and pricing procedures are typically not observed by researchers. Moreover, the empirical conclusions can depend on how these categories are defined. Emekter et al. (2015), for instance, find that default probability generally declines with FICO scores but displays some nonmonotonicity among the higher-score groups, whereas Jagtiani and Lemieux (2019) report a more uniformly decreasing relationship.

Table 1: Comparison of RMSE for $\widehat{g}$ and $\widehat{p}$

	RMSE of $\widehat{g}$		RMSE of $\widehat{p}$	
n	RFP	RF	RFP	RF
Panel A: Sinusoidal				
1,000	0.6681	0.7196	0.1975	0.2089
2,000	0.5805	0.6667	0.1684	0.1888
5,000	0.4767	0.6076	0.1366	0.1671
Panel B: Sigmoid, $\kappa = 1$				
1,000	0.4188	0.5061	0.1169	0.1311
2,000	0.3712	0.4566	0.1027	0.1158
5,000	0.3219	0.4047	0.0883	0.1008
Panel C: Sigmoid, $\kappa = 100$				
1,000	0.4315	0.5235	0.1257	0.1410
2,000	0.3655	0.4579	0.1061	0.1209
5,000	0.2983	0.3799	0.0852	0.0972
Panel D: Threshold				
1,000	0.4341	0.5162	0.1285	0.1428
2,000	0.3687	0.4523	0.1089	0.1227
5,000	0.3034	0.3757	0.0883	0.0993
Panel E: Sparse linear				
1,000	0.4741	0.6378	0.1259	0.1541
2,000	0.4134	0.5825	0.1083	0.1371
5,000	0.3606	0.5197	0.0936	0.1198

Notes: This table reports results from 1,000 Monte Carlo replications with a test-sample size of 10,000. RFP denotes Random Forest Probit and RF denotes the standard random forest regression. Smaller values indicate better performance.

Table 2: Comparison of MAD for $\widehat{g}$ and $\widehat{p}$

	MAD of $\widehat{g}$		MAD of $\widehat{p}$	
n	RFP	RF	RFP	RF
Panel A: Sinusoidal				
1,000	0.5295	0.5729	0.1607	0.1754
2,000	0.4555	0.5272	0.1350	0.1584
5,000	0.3701	0.4770	0.1070	0.1400
Panel B: Sigmoid, $\kappa = 1$				
1,000	0.3309	0.3960	0.0908	0.1076
2,000	0.2926	0.3545	0.0793	0.0946
5,000	0.2532	0.3110	0.0679	0.0816
Panel C: Sigmoid, $\kappa = 100$				
1,000	0.3453	0.4189	0.0972	0.1160
2,000	0.2921	0.3642	0.0813	0.0988
5,000	0.2380	0.2993	0.0652	0.0791
Panel D: Threshold				
1,000	0.3472	0.4133	0.0998	0.1174
2,000	0.2941	0.3599	0.0838	0.1003
5,000	0.2413	0.2960	0.0677	0.0805
Panel E: Sparse linear				
1,000	0.3745	0.5013	0.0966	0.1289
2,000	0.3253	0.4544	0.0826	0.1142
5,000	0.2831	0.4010	0.0709	0.0985

Notes: This table reports results from 1,000 Monte Carlo replications with a test-sample size of 10,000. RFP denotes Random Forest Probit and RF denotes the standard random forest regression. Smaller values indicate better performance.

Table 3: Comparison of RMSE for $\widehat{g}$ and $\widehat{p}$

	RMSE of $\widehat{g}$				RMSE of $\widehat{p}$			
n	RFP	KP	SPP	LP	RFP	KP	SPP	LP
Panel A: Sinusoidal								
1,000	0.6681	0.5279	0.5342	0.9327	0.1975	0.1493	0.1459	0.2857
2,000	0.5805	0.4597	0.4189	0.9278	0.1684	0.1241	0.1175	0.2832
5,000	0.4767	0.3949	0.3524	0.9250	0.1366	0.1023	0.0986	0.2818
Panel B: Sigmoid, $\kappa = 1$								
1,000	0.4188	0.3977	0.4577	0.1598	0.1169	0.0919	0.1167	0.0459
2,000	0.3712	0.3641	0.2993	0.1131	0.1027	0.0785	0.0804	0.0327
5,000	0.3219	0.3419	0.1810	0.0727	0.0883	0.0685	0.0501	0.0211
Panel C: Sigmoid, $\kappa = 100$								
1,000	0.4315	0.5652	0.6409	0.5130	0.1257	0.1637	0.1878	0.1524
2,000	0.3655	0.5414	0.5603	0.5020	0.1061	0.1568	0.1666	0.1487
5,000	0.2983	0.5255	0.5173	0.4954	0.0852	0.1523	0.1536	0.1463
Panel D: Threshold								
1,000	0.4341	0.5580	0.6353	0.5094	0.1285	0.1645	0.1890	0.1536
2,000	0.3687	0.5352	0.5562	0.4985	0.1089	0.1580	0.1678	0.1498
5,000	0.3034	0.5179	0.5139	0.4920	0.0883	0.1533	0.1548	0.1475
Panel E: Sparse linear								
1,000	0.4741	0.6583	1.5856	0.2311	0.1259	0.1540	0.2605	0.0634
2,000	0.4134	0.5935	0.7027	0.1574	0.1083	0.1319	0.1612	0.0439
5,000	0.3606	0.5501	0.3676	0.0987	0.0936	0.1163	0.0953	0.0278

Notes: This table reports results from 1,000 Monte Carlo replications with a test-sample size of 10,000. RFP denotes Random Forest Probit, KP denotes Kernelized Probit, SPB denotes second-order polynomial probit, and LP denotes conventional linear probit. Smaller values indicate better performance.

Table 4: Comparison of MAD for $\widehat{g}$ and $\widehat{p}$

	MAD of $\widehat{g}$				MAD of $\widehat{p}$			
n	RFP	KP	SPP	LP	RFP	KP	SPP	LP
Panel A: Sinusoidal								
1,000	0.5295	0.4114	0.4063	0.7574	0.1607	0.1184	0.1093	0.2458
2,000	0.4555	0.3529	0.3188	0.7538	0.1350	0.0980	0.0879	0.2444
5,000	0.3701	0.2992	0.2681	0.7518	0.1070	0.0792	0.0739	0.2437
Panel B: Sigmoid, $\kappa = 1$								
1,000	0.3309	0.2960	0.3509	0.1259	0.0908	0.0724	0.0864	0.0341
2,000	0.2926	0.2642	0.2304	0.0891	0.0793	0.0618	0.0592	0.0242
5,000	0.2532	0.2402	0.1397	0.0575	0.0679	0.0534	0.0368	0.0157
Panel C: Sigmoid, $\kappa = 100$								
1,000	0.3453	0.4551	0.5123	0.4118	0.0972	0.1302	0.1444	0.1185
2,000	0.2921	0.4363	0.4500	0.4029	0.0813	0.1243	0.1289	0.1158
5,000	0.2380	0.4237	0.4156	0.3976	0.0652	0.1205	0.1194	0.1142
Panel D: Threshold								
1,000	0.3472	0.4494	0.5080	0.4087	0.0998	0.1313	0.1461	0.1200
2,000	0.2941	0.4314	0.4467	0.4000	0.0838	0.1256	0.1305	0.1173
5,000	0.2413	0.4176	0.4127	0.3947	0.0677	0.1213	0.1210	0.1156
Panel E: Sparse linear								
1,000	0.3745	0.4968	1.2240	0.1829	0.0966	0.1265	0.1951	0.0462
2,000	0.3253	0.4354	0.5475	0.1249	0.0826	0.1061	0.1182	0.0320
5,000	0.2831	0.3917	0.2873	0.0783	0.0709	0.0912	0.0692	0.0202

Notes: This table reports results from 1,000 Monte Carlo replications with a test-sample size of 10,000. RFP denotes Random Forest Probit, KP denotes Kernelized Probit, SPB denotes second-order polynomial probit, and LP denotes conventional linear probit. Smaller values indicate better performance.

These findings suggest that the treatment of FICO thresholds is a nontrivial modeling choice.

Following Jagtiani and Lemieux (2019) and Emekter et al. (2015), we use loan-level data from LendingClub. Our sample covers loans originated between 2012 and 2018 and contains 1,213,853 observations. Rather than discretizing the FICO score into researcher-defined categories, we retain the original score and allow the random forest to identify nonlinearities and threshold behavior directly from the data. We consider the binary choice model

$$Y_i^* = g(X_i) - U_i,$$
$$Y_i = 1\{Y_i^* > 0\},$$

where i indexes individual loans, $Y_i = 1$ indicates default, $g(X_i)$ is the systematic component governing default disutility, and U_i is the idiosyncratic error. The vector X_i contains 19 borrower and loan characteristics, including the FICO score, loan amount, loan term, home-ownership status, annual income, verification status, loan purpose, debt-to-income ratio, revolving-credit utilization, credit history, and recent credit inquiries.

We exclude the loan interest rate from X_i because the lender sets the rate using information about the borrower's risk. The rate may therefore reflect both observed borrower characteristics and unobserved factors related to default risk. Controlling for it could therefore remove part of the relationship between borrower characteristics, especially the FICO score, and the underlying propensity to default. Table 5 in Appendix D reports the summary statistics.

We select the tuning parameters of the RFP estimator by stratified 10-fold cross-validation. Let

$$\xi = (m_{\text{try}}, k, n_{\min}, \varphi)$$

denote the collection of tuning parameters. With $d = 19$ covariates, the candidate values for m_{try} are $m_{\text{try}} \in \{4, 6, 19\}$, corresponding approximately to $\sqrt{d}$, $d/3$, and d. The candidate maximum depths are $k \in \{10, 20, 30, 40, \infty\}$, where $k = \infty$ denotes an unrestricted maximum depth. The minimum terminal-node size is selected from $n_{\min} \in \{5, 25, 50, 100, 200, 500\}$, and the pseudo-count parameter is selected from $\varphi \in \{0.001, 0.01, 0.1, 1, 2\}$.

Each candidate RFP contains 1,000 trees. After the optimal tuning parameters are selected, we refit the RFP estimator on the full sample of 1,213,853 loans using the selected $(m_{\text{try}}, k, n_{\min}, \varphi)$ and 1,000 trees. The resulting full-sample estimator is then used for the subsequent interpretation of the relationship between FICO scores and the systematic component of default decisions. The tuning-parameter selection yields the optimal hyperparameters given by $(m_{\text{try}}, k, n_{\min}, \varphi) = (4, \infty, 5, 2)$.

To examine how the estimated systematic component varies with the FICO score, we use the Individual Conditional Expectation (ICE) curve proposed by Goldstein

et al. (2015). For covariate j and a borrower characterized by $X_{-j} = x_{-j}$, the ICE curve is

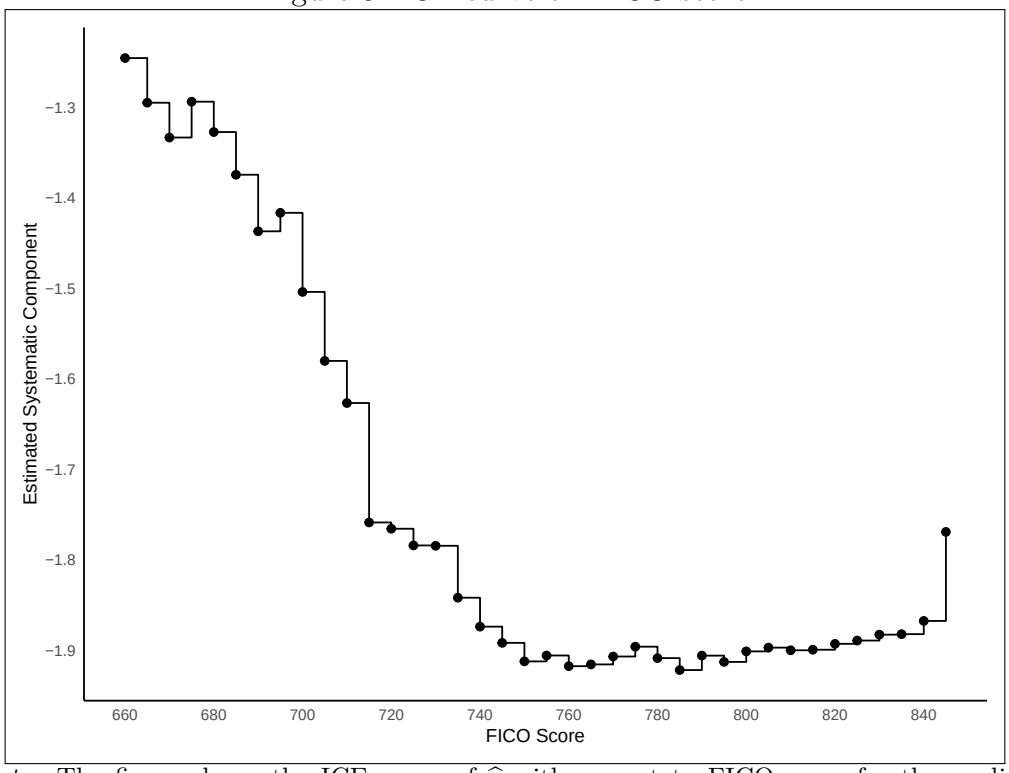
$$\widehat{\mathrm{ICE}}_j(z; x_{-j}) = \widehat{g}(X_j = z, X_{-j} = x_{-j}).$$

Unlike a partial dependence curve, which averages over the distribution of the remaining covariates, an ICE curve holds X_{-j} fixed and therefore traces the heterogenuous effect of X_j on the fitted systematic component for a particular covariate profile.

We set X_j equal to the FICO score and construct a representative borrower by setting continuous covariates to their sample medians and categorical covariates to their sample modes. Figure 3 reports the resulting ICE curve. The estimated systematic component generally declines as the FICO score increases, indicating that higher credit scores are associated with a lower utility to default for this representative borrower. Importantly, however, the relationship is highly nonlinear and distinctly step-shaped rather than smooth.

The most pronounced declines occur over the middle portion of the FICO distribution. The curve exhibits several sharp downward movements between approximately 700 and 715, followed by additional decreases through the 730s and early 740s. By around 750, the estimated systematic component has fallen substantially relative to its level at lower FICO scores and subsequently becomes nearly flat over a wide range of higher scores. These patterns indicate that the marginal association between additional improvements in FICO and the latent default disutility is considerably stronger over some score ranges than over others. The stepwise behavior is consistent with the presence of threshold-type heterogeneity that would be difficult to capture using a smooth linear specification or a small number of prespecified FICO categories. At the extreme upper end of the FICO range, the ICE curve turns upward, particularly at the highest observed score values. We do not interpret this finding as relatively few observations are near the boundary of the covariate support.

Figure 3: ICE curve on FICO score



Note: The figure shows the ICE curve of $\widehat{g}$ with respect to FICO score for the median borrower.

7 Conclusion

This paper adapts random forests to estimate binary choice models without imposing a smooth or parametric functional form on the systematic component. The proposed estimator transforms adjusted within-leaf sample proportion through the inverse of the known error distribution and averages the resulting tree-level estimates of the systematic component. Profiling out the leaf coefficients preserves the standard CART splitting rules, allowing the procedure to use existing random forest implementations without additional numerical optimization at each split. The proposed method introduces additional tuning parameters, which can be selected jointly with the standard random forest tuning parameters through stratified cross-validation. Under suitable regularity conditions, the theoretical analysis establishes L^2 convergence rates while accommodating continuous and discrete covariates whose dimensions may grow polynomially with the sample size.

The simulations identify the settings in which this approach is most useful. Our proposed RFP estimator achieves lower RMSE and MAD than standard random for-

est regression for both the systematic component and the choice probability across all reported designs and sample sizes, which support adapting random forest estimation to the binary choice structure. Among the competing probit estimators, its advantage is strongest when the systematic component exhibits sharp nonlinearities or discontinuities. The LendingClub application demonstrates how the estimator can describe heterogeneous and nonlinear effects in the systematic component without imposing prespecified FICO categories.

Three extensions would broaden the scope of the approach. The first is joint estimation of the systematic component and an unknown error distribution. The second is to extend our methodology to multinomial choice models. The third is to develop comparable estimation procedures for boosted-tree methods. These extensions would provide further ways to combine flexible tree-based methods with the economic structure of discrete choice models.

Remark on Notation Throughout the proofs, we will use C to denote a generic positive constant whose value may change from line to line. For an arbitrary set A, denote its complement by A^c and its cardinality by $|A|$. Let "Δ" be the symmetric difference of two sets. For two numerical sequences a_n and b_n, write $a_n \sim b_n$ if $a_n/b_n \to 1$ as $n \to \infty$. Finally, we will use "$\prod$" to denote both Cartesian product and numerical product, depending on the contexts.

A Grid Technique

In this section, we introduce a grid technique, following Chi et al. (2022) with modifications that allow for discrete-type covariates. Recall that $X \in [0,1]^d$. Without loss of generality (wlog), let $\{1, \cdots, d_1\}$ and $\{d_1+1, \cdots, d\}$ be the index set for the continuous-type and discrete-type covariates, respectively. Let a constant $\rho > 0$ be given.

For $j \in \{1, \cdots, d_1\}$, i.e., along the j-th coordinate corresponding to a continuous covariate, we discretize the unit interval $[0,1]$ by a set of grid points defined as

$$G_j = \left\{ \frac{m}{\lceil n^{1+\rho} \rceil} : m = 0, 1, \cdots, \lceil n^{1+\rho} \rceil \right\}.$$

For $j \in \{1, \cdots, d_1\}$, the set of grid points G_j partitions the j-th unit interval into a disjoint union given by[24]

$$[0,1] = \bigcup_{m=1}^{\lceil n^{1+\rho} \rceil - 1} \left[\frac{m-1}{\lceil n^{1+\rho} \rceil}, \frac{m}{\lceil n^{1+\rho} \rceil} \right) \bigcup \left[\frac{\lceil n^{1+\rho} \rceil - 1}{\lceil n^{1+\rho} \rceil}, 1 \right].$$

For $j \in \{d_1+1, \cdots, d\}$, i.e., along the coordinate corresponding to a discrete covariate, the grid points are given by

$$G_j = \{0,1\} \cup \{c_{j,1}, \cdots, c_{j,S_j}\},$$

where $\{c_{j,1}, \cdots, c_{j,S_j}\} = \text{supp}(X_j)$ is a finite set under Assumption 3(ii), i.e., $S_j = |\text{supp}(X_j)| < \infty$, and $\text{supp}(X_j)$ denotes the support of X_j. For $j \in \{d_1+1, \cdots, d\}$, G_j partitions the j-th unit interval into a disjoint union given by[25]

$$[0,1] = [0, c_{j,1}) \cup \left(\bigcup_{m=1}^{S_j - 1} [c_{j,m}, c_{j,m+1}) \right) \cup [c_{j,S_j}, 1].$$

[24] Alternatively, we may, wlog, write $\bigcup_{m=1}^{\lceil n^{1+\rho} \rceil} \left[\frac{m-1}{\lceil n^{1+\rho} \rceil}, \frac{m}{\lceil n^{1+\rho} \rceil} \right)$, since by Assumption 4, X_j has a density with respect to the Lebesgue measure.

[25] For the same reason as in the above footnote, we may alternatively, remove the first intervel $[0, c_{j,1})$ wlog.

It can be seen that these grid points cut $[0,1]^d$ into $|G_1|^{d_1} \times \prod_{j=d_1+1}^{d} |G_j|$ number of little hyperrectangles, which only depend on the sample size n and the fineness parameter $\rho > 0$ and are independent of the dataset $\mathcal{D}$.

For a cell t, define its approximating cell $t^\#$ by moving all boundaries of t to the nearest grid lines given by the grid points. Similarly, for a partition T, define $T^\# = \{t^\# : t \in T\}$. For example, suppose $d = 1$, $X \in [0,1]$ is continuous, $n = 2$, and $\rho = 1$. Then the set of grid points is $\{0, 1/4, 1/2, 3/4, 1\}$ and the grid is $\{[0,1/4), [1/4,1/2), [1/2,3/4), [3/4,1)\}$. If $t = [0,1/8)$, then $t^\# = [0,1/4)$. If $t = [0.46, 0.9)$, then $t^\# = [1/2,1)$. If X is discrete, then $t = t^\#$ trivially. To be rigorous, below we provide the mathematical definition of $t^\#$.

For an interval $I_j = [L_j, R_j)$, define

$$L_j^\# = \min\{\arg\min_{g \in G_j} |g - L_j|\} \quad \text{and} \quad R_j^\# = \min\{\arg\min_{g \in G_j} |g - R_j|\},$$

and $I^\# = [L_j^\#, R_j^\#)$[26][27]. For $I_j = [L_j, 1]$, define $I_j^\# = [L_j^\#, 1]$. Then, for a cell $t = \prod_{j=1}^d I_j$, define $t^\# = \prod_{j=1}^d I_j^\#$.

It should be emphasized that by construction, the sharp notation have two important properties. First, if $t = t' \cup t''$ and t', t'' are disjoint, then $t^\# = (t^\#)' \cup (t^\#)''$ and $(t')^\#, (t'')^\#$ are also disjoint, i.e., if t' and t'' are the child cells of t, then $(t^\#)'$ and $(t^\#)''$ are the child cells of $t^\#$. Second, if a set of pairwise disjoint t's forms a partition T of $[0,1]^d$, then $t^\#$'s are also pairwise disjoint and form the partition $T^\#$ of $[0,1]^d$.

For $j \in \{1, \cdots, d_1\}$ and $0 \le m \le \lceil n^{1+\rho_1} \rceil - 1$, define the strip

$$t_0(j,m) = [0,1]^{j-1} \times \left[\frac{m-1}{\lceil n^{1+\rho_1} \rceil}, \frac{m}{\lceil n^{1+\rho_1} \rceil} \right) \times [0,1]^{d-j}, \quad 1 \le m \le \lceil n^{1+\rho_1} \rceil - 1 \quad \text{and}$$
$$t_0(j, \lceil n^{1+\rho_1} \rceil) = [0,1]^{j-1} \times \left[\frac{\lceil n^{1+\rho_1} \rceil - 1}{\lceil n^{1+\rho_1} \rceil}, 1 \right] \times [0,1]^{d-j}.$$

For $j \in \{d_1 + 1, \cdots, d\}$ and any m, simply let $t_0(j,m) = \emptyset$.

An immediate result that justifies the usage of $t^\#$ to approximate t is that for all $1 \le j \le d$, the intervals $I_j \backslash I_j^\#$ and $I_j^\# \backslash I_j$ are contained in $t_0(j,m)$ for some m. Since under Assumption 4,

$$\mathbb{P}\{X \in t_0(j,m)\} = \mathbb{P}\left\{ X_j \in \left[\frac{m-1}{\lceil n^{1+\rho} \rceil}, \frac{m}{\lceil n^{1+\rho} \rceil} \right) \right\} \le \frac{\sup f}{n^{1+\rho}} \tag{44}$$

[26] We set the convention that whenever there are two grid lines that are equally close to a boundary, we shift the boundary to the one associated with the smaller grid point. This convention is captured by the outer "min". Changing "min" to "max" will make no difference in our theoretical analysis.

[27] It is possible to have $L_j^\# = R_j^\#$, in which case $I_j^\# = \emptyset$.

for $1 \le j \le d_1$ and $1 \le m \le \lceil n^{1+\rho} \rceil$, we have

$$\begin{aligned} \mathbb{P}\{X_j \in I_j \Delta I_j^{\#}\} &\le \mathbb{P}\{X_j \in I_j \backslash I_j^{\#}\} + \mathbb{P}\{X_j \in I_j \backslash I_j^{\#}\} \\ &\le 2\mathbb{P}\left\{X \in t_0(j,m)\right\} \le \frac{2 \sup f}{n^{1+\rho}}, \end{aligned} \tag{45}$$

that is, the difference between I_j and $I_j^{\#}$ is negligible in probabilistic sense at a polynomial order. For $d_1 + 1 \le j \le d$, we have $\mathbb{P}\{X \in I_j \Delta I_j^{\#}\} = 0$, and thus, (45) holds for all $1 \le j \le d$.

Furthermore, define $\mathcal{G}_k$ to be the collection of cells that can be made by at most k splits and $\mathcal{G}^{\#}(k)$ to be the collection of the approximating cells $t^{\#}$ for which $t \in \mathcal{G}_k$, that is,

$$\begin{aligned} &\mathcal{G}_k = \{t : t \text{ is a cell made by at most } k \text{ splits}\} \quad \text{and} \\ &\mathcal{G}_k^{\#} = \{t^{\#} : t \in \mathcal{G}_k\} \end{aligned} \tag{46}$$

Cells that have more than k boundaries being split do not belong to $\mathcal{G}_k$. If $k \ge d$, then every cell in $[0,1]^d$ is included in $\mathcal{G}_k$ because every cell in $[0,1]^d$ can be made by no more than d splits.

Finally, we define the sets of large cells and negaligible cells in $\mathcal{G}_k^{\#}$ respectively as

$$\begin{aligned} &\mathcal{G}_{k,+}^{\#} = \{t^{\#} \in \mathcal{G}_k^{\#} : \mathbb{P}\{X \in t\} \ge n^{-1} \log n\} \quad \text{and} \\ &\mathcal{G}_{k,-}^{\#} = \{t^{\#} \in \mathcal{G}_k^{\#} : \mathbb{P}\{X \in t\} < n^{-1} \log n\}. \end{aligned} \tag{47}$$

Notice that by definition, $\mathcal{G}_{k,-}^{\#}$ and $\mathcal{G}_{k,-}^{\#}$ are disjoint and we have $\mathcal{G}_{k,+}^{\#} \cup \mathcal{G}_{k,-}^{\#} = \mathcal{G}_k^{\#} \subset \mathcal{G}_k$.

A fundamental problem in proving random forests consistency is that it involves uncountably many data-dependent T's, which makes it impossible to uniformly bound the estimation errors defined in Lemma 4 for every possible cells. The grid approach provides a solution to such a problem, and allows us to apply the classical Bernstein's inequality to uniformly bound the probabilities for every cell on the grid hyperplane. We summarize some useful results for the grid approach in Lemma 2 and Lemma 3 below. These results will be employed throughout the proofs of the present paper.

Essentially, Lemma 2 below shows that although $\mathcal{G}_k$ is generally uncountable for all $k \ge 1$, $\mathcal{G}_k^{\#}$ becomes a finite set.

Lemma 2. *Let Assumption 3 hold and let the tree depth k satisfy $k = O(\log_\delta n)$ with $\delta > 1$. Then, it holds for all $n \in \mathbb{N}$ that*

$$|\mathcal{G}_k| \le \exp\left(C(\log_\delta n)^2\right).$$

Proof. Note that each split involves choosing one grid line from a total of $d_1(\lceil n^{1+\rho} \rceil + 1) + \sum_{j=d_1+1}^{d} |G_j|$ grid lines, that we can split at most k times, and that k splits can generate up to 2^k distinct cells. Therefore, by Assumption 3, since $k = O(\log_2 n)$, we can deduce that

$$\begin{aligned} |\mathcal{G}_k| &\leq 2^k \left(d_1(\lceil n^{1+\rho} \rceil + 1) + \sum_{j=d_1+1}^{d} (S_j + 2) \right)^k \\ &\leq n^{C \log_\delta 2} \left(C n^{1+\rho+C_1} + C n^{C_2} \right)^{C \log_\delta n} \\ &\leq n^{C \log_\delta 2} \left(C n^{1+\rho+C_1+C_2} \right)^{C \log_\delta n} \end{aligned} \tag{48}$$

Taking logarithms,

$$\begin{aligned} \log (48) &= C \log_\delta 2 \log n + C \log_2 n \left(\log C + (1 + \rho + C_1 + C_2) \log n \right) \\ &\leq C (\log_\delta n)^2, \end{aligned}$$

from which the stated result follows. This completes the proof. □

Of course, by shifting a cell t to $t^\#$, certain errors will be created. However, using the result in (45), we can show that such errors are negligible whose order is governed by the sample size as well as the fineness parameter ρ. We formalize such results in Lemma 3 below.

Lemma 3. *Let Assumption 2 and 4 hold. Define an event E as*

$$E = \bigcap_{j,m} \left\{ \frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t_0(j,m)\} < n^{-1} \log n \right\}, \tag{49}$$

where the intersection is taken over all $1 \leq j \leq d_1$ and $1 \leq m \leq \lceil n^{1+\rho_1} \rceil$.

(i) We have for all $t \in \mathcal{G}_k$ and $n \in \mathbb{N}$,

$$\mathbb{P}\{X \in t \Delta t^\#\} \leq 2k \frac{\sup f}{n^{1+\rho}}.$$

(ii) On event E, it holds that for all $t \in \mathcal{G}_k$ and $n \in \mathbb{N}$,

$$\frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t \Delta t^\#\} \leq 2k \frac{\log n}{n}.$$

(iii) We have for all $n \in \mathbb{N}$ that

$$\mathbb{P}(E^c) \leq \max(1, \sup f) d_1 \lceil n^{1+\rho} \rceil \left(\frac{\sup f}{n^\rho} \right)^{\log n}.$$

Proof. We first prove (i). Since each cell t made by k splits has at most k boundaries not on the grid lines and since $t \backslash t^{\#} = \emptyset$ if all cuts correspond to the discrete-type covariates, it follows that $t \Delta t^{\#}$ is a subset of a union of at most k strips $t_0(j, m)$'s. Hence, we have for any cell t constructed by at most k splits that

$$t \backslash t^{\#} \subset \bigcup_{\ell=1}^{k} t_0(j_\ell, m_\ell),$$

for some $t_0(j_\ell, m_\ell)$'s, where j_ℓ indicates the coordinate chosen at the ℓ-th split and m_ℓ is associated with the grid line. Then, we can conclude that for all t made by at most k splits,

$$\begin{aligned} \mathbb{P}\{X \in t \backslash t^{\#}\} &\leq \mathbb{P}\left\{X \in \bigcup_{\ell=1}^{k} t_0(j_\ell, m_\ell)\right\} \\ &\leq k \max_{1 \leq \ell \leq k} \mathbb{P}\left\{X \in t_0(j_\ell, m_\ell)\right\} \\ &\leq k \frac{\sup f}{n^{1+\rho}}, \end{aligned}$$

where the last inequality follows from (44). By symmetry, we have $\mathbb{P}\{X \in t^{\#} \backslash t\} \leq k(\sup f) n^{-1-\rho}$ as well. Therefore,

$$\begin{aligned} \mathbb{P}\{X \in t \Delta t^{\#}\} &= \mathbb{P}\{X \in t \backslash t^{\#} \cup t^{\#} \backslash t\} \\ &\leq \mathbb{P}\{X \in t \backslash t^{\#}\} + \mathbb{P}\{X \in t^{\#} \backslash t\} \leq 2k \frac{\sup f}{n^{1+\rho}}, \end{aligned}$$

as was to be shown.

Next, we prove (ii). Similar to the arguments in the proof of (i), we have on event E, for every cell t constructed by at most k splits,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t \Delta t^{\#}\} &\leq 2k \left(\max_{1 \leq \ell \leq k} \frac{1}{n} \sum_{i=1}^{n} 1\left\{X_i \in t_0(j_\ell, m_\ell)\right\} \right) \\ &\leq 2k \frac{\log n}{n}, \end{aligned}$$

which proves (ii).

Finally, we prove (iii). Write $\ell_0 = \lfloor \log n \rfloor$. By the definition of E given in (49),

we can deduce that

$$\begin{aligned}\mathbb{P}\left(\mathcal{A}^c\right) &\leq \sum_{j,m} \mathbb{P}\left\{\sum_{i=1}^{n} 1\left\{x_i \in t_0(j,m)\right\} \geq \ell_0\right\} \\ &\leq d_1\lceil n^{1+\rho}\rceil \left(\sup_{j,m} \mathbb{P}\left\{\sum_{i=1}^{n} 1\left\{X_i \in t_0(j,m)\right\} \geq \ell_0\right\}\right) \\ &= d_1\lceil n^{1+\rho}\rceil \sup_{j,m}\left(\sum_{\ell=\ell_0}^{n}\binom{n}{\ell}\mathbb{P}\left\{X \in t_0(j,m)\right\}^{\ell}\left(1-\mathbb{P}\left\{X \in t_0(j,m)\right\}\right)^{n-\ell}\right),\end{aligned} \tag{50}$$

where the equality holds due to Assumption 2 and due to the fact that $\sum_{i=1}^{n} 1\left\{X_i \in t_0(j,k)\right\}$ is binomially distributed with parameters n and $\mathbb{P}\left\{X \in t_0(j,m)\right\}$.

For n such that $\frac{\sup f}{n^{1+\rho}} < 1$, or equivalently, $n > (\sup f)^{\frac{1}{1+\rho}}$, we have

$$\begin{aligned}(50) &\leq d_1\lceil n^{1+\rho}\rceil \sum_{\ell=\ell_0}^{n}\binom{n}{\ell}\left(\frac{\sup f}{n^{1+\rho}}\right)^{\ell}\left(1-\frac{\sup f}{n^{1+\rho}}\right)^{n-\ell} \\ &\leq d_1\lceil n^{1+\rho}\rceil \sum_{\ell=\ell_0}^{n}\binom{n}{\ell}\left(\frac{\sup f}{n^{1+\rho}}\right)^{\ell}\end{aligned} \tag{51}$$

where the first inequality holds because of (44) and because the complement of binomial cumulative distribution function is increasing in the probability of success. On the other hand, if $\frac{\sup f}{n^{1+\rho}} \geq 1$, then (51) holds trivially.

Thus, straightforward calculations yield that

$$\begin{aligned}(51) &\leq d_1\lceil n^{1+\rho}\rceil \sum_{\ell=\ell_0}^{n}\frac{1}{\ell!}\left(\frac{\sup f}{n^{\rho}}\right)^{\ell} \\ &\leq d_1\lceil n^{1+\rho}\rceil \frac{1}{\ell_0!}\sum_{\ell=\ell_0}^{\infty}\left(\frac{\sup f}{n^{\rho}}\right)^{\ell} \\ &\leq d_1\lceil n^{1+\rho}\rceil \left(\frac{\sup f}{n^{\rho}}\right)^{\ell_0}\frac{1}{\ell_0!}\frac{1}{1-\frac{\sup f}{n^{\rho}}} \\ &\leq d_1\lceil n^{1+\rho}\rceil \left(\frac{\sup f}{n^{\rho}}\right)^{\ell_0}\end{aligned}$$

for all n such that $n > (\sup f)^{\frac{1}{1+\rho}}$ and

$$\begin{aligned}(51) &\leq d_1\lceil n^{1+\rho}\rceil \sum_{\ell=\ell_0}^{n} \frac{1}{\ell!}\left(\frac{\sup f}{n^\rho}\right)^\ell \\ &\leq d_1\lceil n^{1+\rho}\rceil n\left(\frac{\sup f}{n^\rho}\right)^\ell \\ &\leq (\sup f) d_1\lceil n^{1+\rho}\rceil \left(\frac{\sup f}{n^\rho}\right)^{\ell_0}\end{aligned}$$

for all n such that $n \leq (\sup f)^{\frac{1}{1+\rho}}$, thus proving (iii). This completes the proof of Lemma 3. □

B Proofs of Useful Lemmas

B.1 Proof of Lemma 1

Proof. We need to show that

$$\begin{aligned}&\mathbb{P}\{X \in t\}\mathrm{var}(Y|X \in t) \\ &= \mathbb{P}\{X \in t'\}\mathrm{var}(Y|X \in t') + \mathbb{P}\{X \in t''\}\mathrm{var}(Y|X \in t'') \\ &\quad + \mathbb{P}\{X \in t'\}\left(\mathbb{E}(Y|X \in t') - \mathbb{E}(Y|X \in t)\right)^2 + \mathbb{P}\{X \in t''\}\left(\mathbb{E}(Y|X \in t'') - \mathbb{E}(Y|X \in t)\right)^2.\end{aligned}$$

Notice that

$$\begin{aligned}&\mathbb{P}\{X \in t\}\mathrm{var}(Y|X \in t) \\ &= \mathbb{P}\{X \in t\}\mathbb{E}(Y^2|X \in t) - \mathbb{P}\{X \in t\}\mathbb{E}(Y|X \in t)^2 \\ &= \mathbb{P}\{X \in t\}\mathbb{E}(Y^2|X \in t) - \left(\mathbb{P}\{X \in t'\}\mathbb{E}(Y|X \in t')^2 + \mathbb{P}\{X \in t''\}\mathbb{E}(Y|X \in t'')^2\right) \qquad (52) \\ &\quad + \left(\mathbb{P}\{X \in t'\}\mathbb{E}(Y|X \in t')^2 + \mathbb{P}\{X \in t''\}\mathbb{E}(Y|X \in t'')^2\right) - \mathbb{P}\{X \in t\}\mathbb{E}(Y^2 1\{X \in t\}). \qquad (53)\end{aligned}$$

We have

$$\begin{aligned}(52) &= \mathbb{E}(Y^2(1\{X \in t'\} + 1\{X \in t''\})) - \mathbb{P}\{X \in t'\}\mathbb{E}(Y|X \in t')^2 - \mathbb{P}\{X \in t''\}\mathbb{E}(Y|X \in t'')^2 \\ &= \mathbb{P}\{X \in t'\}\mathbb{E}(Y^2|X \in t') + \mathbb{P}\{X \in t''\}\mathbb{E}(Y^2|X \in t'') \\ &\quad - \mathbb{P}\{X \in t'\}\mathbb{E}(Y|X \in t')^2 - \mathbb{P}\{X \in t''\}\mathbb{E}(Y|X \in t'')^2 \\ &= \mathbb{P}\{X \in t'\}\mathrm{var}(Y|X \in t') + \mathbb{P}\{X \in t''\}\mathrm{var}(Y|X \in t'')\end{aligned}$$

and

$$
\begin{aligned}
(53) &= \mathbb{P}\{X \in t'\}\mathbb{E}(Y|X \in t')^2 + \mathbb{P}\{X \in t''\}\mathbb{E}(Y|X \in t'')^2 + \mathbb{P}\{X \in t\}\mathbb{E}(Y|X \in t)^2 \\
&\quad - 2\Big(\mathbb{P}\{X \in t\}\mathbb{E}(Y|X \in t)\Big)\mathbb{E}(Y|X \in t) \\
&= \mathbb{P}\{X \in t'\}\mathbb{E}(Y|X \in t')^2 + \mathbb{P}\{X \in t''\}\mathbb{E}(Y|X \in t'')^2 + (\mathbb{P}\{X \in t'\} + \mathbb{P}\{X \in t''\})\mathbb{E}(Y|X \in t)^2 \\
&\quad - 2\Big(\mathbb{P}\{X \in t'\}\mathbb{E}(Y|X \in t') + \mathbb{P}\{X \in t''\}\mathbb{E}(Y|X \in t')\Big)\mathbb{E}(Y|X \in t) \\
&= \mathbb{P}\{X \in t'\}\left(\mathbb{E}(Y|X \in t') - \mathbb{E}(Y|X \in t)\right)^2 + \mathbb{P}\{X \in t''\}\left(\mathbb{E}(Y|X \in t'') - \mathbb{E}(Y|X \in t)\right)^2,
\end{aligned}
$$

as was to be shown. This completes the proof. □

Lemma 4. *Let Assumption 2 and 3 hold. Let $\mathcal{G}_k$, $\mathcal{G}_k^{\#}$, and $\mathcal{G}_{k,+}^{\#}$ be given as in (46) and (47). Let $k = O(\log_\delta n)$ with $\delta > 1$ be given. Define*

$$
\begin{aligned}
A_{k,1} &= \bigcap_{t \in \mathcal{G}_{k,+}^{\#}} \left\{ \left| \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\} - \mathbb{P}\{X \in t\} \right| < n^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} (\log\log n) \sqrt{\mathbb{P}\{X \in t\}} \right\}, \\
A_{k,2} &= \bigcap_{t \in \mathcal{G}_{k,+}^{\#}} \left\{ \left| \frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\} - \mathbb{E}(Y 1\{X \in t\}) \right| < n^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} (\log\log n) \sqrt{\mathbb{P}\{X \in t\}} \right\}, \\
B_k &= \bigcap_{t \in \mathcal{G}_{k,-}^{\#}} \left\{ \left| \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\} - \mathbb{P}\{X \in t\} \right| < n^{-1} (\log n)^2 \log\log n \right\}.
\end{aligned}
$$

Then, it holds that

$$
\mathbb{P}\Big(A_{k,1}^c\Big) \le 2\exp\left(-C(\log n)^2 \log\log n\right), \tag{54}
$$

$$
\mathbb{P}\Big(A_{k,2}^c\Big) \le 2\exp\left(-C(\log n)^2 \log\log n\right), \text{ and} \tag{55}
$$

$$
\mathbb{P}\Big(B_k^c\Big) \le 2\exp\left(-C(\log n)^2 \log\log n\right) \tag{56}
$$

for all $n \ge 2$.

Proof. We will proceed in the order (54), (56), and (55).

First, we establish (54). Note that for every cell $t \subset [0,1]^d$, $0 \le 1\{X_i \in t\} \le 1$ and $\mathbb{E}1\{X_i \in t\} = \mathbb{P}\{X \in t\}$ for all i, by Assumption 2. Then, we can apply Bernstein's inequality to $1\{X_i \in t\}$ and find that for any $\epsilon > 0$,

$$
\mathbb{P}\left\{ \left| \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\} - \mathbb{P}\{X \in t\} \right| \ge \epsilon \right\} \le 2\exp\left(-\frac{n\epsilon^2/2}{(1 - \mathbb{P}\{X \in t\})\mathbb{P}\{X \in t\} + \epsilon/3} \right).
$$

Setting $\epsilon = n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}(\log\log n)\sqrt{\mathbb{P}\{X \in t\}}$, we have,

$$\begin{aligned}
&\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\} - \mathbb{P}\{X \in t\}\right| \geq n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}(\log\log n)\sqrt{\mathbb{P}\{X \in t\}}\right\} \\
&\leq 2\exp\left(\frac{-\frac{1}{2}(\log n)^3(\log\log n)^2}{1 - \mathbb{P}\{X \in t\} + \frac{1}{3}n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}(\log\log n)(\mathbb{P}\{X \in t\})^{-\frac{1}{2}}}\right) \\
&\leq 2\exp\left(\frac{-\frac{1}{2}(\log n)^3(\log\log n)^2}{1 + \frac{1}{3}(\log n)\log\log n}\right) \\
&\leq 2\exp\Big(-C(\log n)^2\log\log n\Big)
\end{aligned}$$

uniformly in $\mathcal{G}_{k,+}^{\#}$, where the second inequality holds since $\mathbb{P}\{X \in t\} \geq n^{-1}\log n$ for $t \in \mathcal{G}_{k,+}^{\#}$. This, together with Lemma 2, ensures that

$$\begin{aligned}
\mathbb{P}\Big(A_{k,1}^c\Big) &\leq \sum_{t \in \mathcal{G}_{k,+}^{\#}} \mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} 1\{x_i \in t\} - \mathbb{P}\{X \in t\}\right| \geq n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}(\log\log n)\sqrt{\mathbb{P}\{X \in t\}}\right\} \\
&\leq \exp\Big(C(\log_\delta n)^2\Big) \times 2\exp\Big(-C(\log n)^2\log\log n\Big) \\
&\leq 2\exp\Big(-C(\log n)^2\log\log n\Big)
\end{aligned} \tag{57}$$

for all $n \geq 2$. This gives (54).

Next, to establish (56), similar to (54), we apply Bernstein's inequality to $1\{X_i \in t\}$ with $\epsilon = n^{-1}(\log n)^2\log\log n$ to get

$$\begin{aligned}
&\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\} - \mathbb{P}\{X \in t\}\right| \geq n^{-1}(\log n)^2\log\log n\right\} \\
&\leq 2\exp\left(\frac{-\frac{1}{2}n^{-1}(\log n)^4(\log\log n)^2}{n^{-1}\log n + \frac{1}{3}n^{-1}(\log n)^2\log\log n}\right) \\
&\leq 2\exp\Big(-C(\log n)^2\log\log n\Big)
\end{aligned}$$

uniformly in $\mathcal{G}_{k,-}^{\#}$, where the second inequality follows from the definition of $\mathcal{G}_{k,-}^{\#}$. Now, proceeding exactly as in (57) proves (56).

Finally, we prove (55). Note that $0 \leq 1\{X_i \in t\}Y_i \leq 1$ and $\mathbb{E}\Big(1\{X_i \in t\}Y_i\Big) = \mathbb{E}\Big(1\{X \in t\}p(X)\Big)$ by Assumption 2. Moreover,

$$\mathrm{var}\Big(1\{X_i \in t\}Y_i\Big) \leq \mathbb{E}(Y^2 1\{X \in t\}) \leq \mathbb{P}\{X \in t\}. \tag{58}$$

Then, applying Bernstein's inequality with $\epsilon = n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}(\log\log n)\sqrt{\mathbb{P}(X \in t)}$ yields

$$\begin{aligned}
&\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\}Y_i - \mathbb{E}\Big(1\{X \in t\}Y\Big)\right| \geq n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}(\log\log n)\sqrt{\mathbb{P}(X \in t)}\right\} \\
&\leq 2\exp\left(-\frac{\frac{1}{2}(\log n)^3(\log\log n)^2\mathbb{P}\{X \in t\}}{\mathbb{P}\{X \in t\} + \frac{1}{3}n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}(\log\log n)\sqrt{\mathbb{P}\{X \in t\}}}\right) \\
&\leq 2\exp\left(-\frac{\frac{1}{2}(\log n)^3(\log\log n)^2}{1 + \frac{1}{3}(\log n)(\log\log n)}\right) \\
&\leq 2\exp\Big(-C(\log n)^2(\log\log n)\Big)
\end{aligned}$$

for all $n \geq 2$, uniformly in $\mathcal{G}_{k,+}^{\#}$. Then, following the steps as in (57), we can establish (55). This completes the proof of Lemma 4. □

Lemma 5. *Let Assumption 2, 3, and 4 hold. Let the events $A_{1,k}$, $A_{2,k}$, B_k and E be given as in Lemma 4 and (49). Let $k = O(\log_\delta n)$ with $\delta > 1$ be given. Then, it holds that for all $t \in \mathcal{G}_k$, if $\mathbb{P}\{X \in t\} \leq n^{-1}\log n$, then on $B_k \cap E$,*

$$\left|\frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\} - \mathbb{P}\{X \in t\}\right| \leq Cn^{-1}(\log n)^2\log\log n; \tag{59}$$

if otherwise $\mathbb{P}\{X \in t\} > n^{-1}\log n$, then

$$\mathbb{P}\{X \in t\}^{-\frac{1}{2}}\left|\frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\} - \mathbb{P}\{X \in t\}\right| \leq Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n \tag{60}$$

on $A_{1,k} \cap E$ and

$$\mathbb{P}\{X \in t\}^{-\frac{1}{2}}\left|\frac{1}{n}\sum_{i=1}^{n} Y_i 1\{X_i \in t\} - \mathbb{E}(Y1\{X \in t\})\right| \leq Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n. \tag{61}$$

on $A_{2,k} \cap E$.

Proof. The key idea of the proof is to decompose the sample-averaging error into three components. The first component captures the discrepancy arising from shifting the sample analogue to its counterpart on the grid hyperplane. The second corresponds to the original error restricted to the grid hyperplane. The third accounts for the discrepancy induced by shifting the population quantity onto its grid-hyperplane counterpart. Lemma 3 provides the main technical tool for controlling these three components.

We first establish (59). As discussed above, the LHS of (59) is decomposed as

$$\left|\frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t\}-\mathbb{P}\{X\in t\}\right| \le \left|\frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t\}-\frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t^{\#}\}\right| \tag{62}$$

$$+\left|\frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t^{\#}\}-\mathbb{P}\{X\in t^{\#}\}\right| \tag{63}$$

$$+\left|\mathbb{P}\{X\in t^{\#}\}-\mathbb{P}\{X\in t\}\right| \tag{64}$$

For the RHS of (62), we have for all $t\in\mathcal{G}_k$, on event E,

$$\begin{aligned}\left|\frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t\}-\frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t^{\#}\}\right| &\le \frac{1}{n}\sum_{i=1}^{n}|1\{X_i\in t\}-1\{X_i\in t^{\#}\}|\\ &= \frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t\Delta t^{\#}\}\\ &\le 2kn^{-1}\log n\end{aligned}$$

where the last inequality is due to Lemma 3(ii). For (63), we have for all $t\in\mathcal{G}_k$ such that $\mathbb{P}\{X\in t\}\le n^{-1}\log n$, on B_k,

$$(63)\le n^{-1}(\log n)^2\log\log n.$$

For (64), by Lemma 3(i), for all $t\in\mathcal{G}_k$, it holds that

$$\begin{aligned}\left|\mathbb{P}\{X\in t^{\#}\}-\mathbb{P}\{X\in t\}\right| &\le \mathbb{E}|1\{X\in t\}-1\{X\in t^{\#}\}|\\ &= \mathbb{P}\{X\in t\Delta t^{\#}\}\\ &\le 2k(\sup f)n^{-1-\rho}\end{aligned}$$

for any given $\rho>0$. Thus, combining the results for the three items, together with the condition that $k=O(\log_\delta n)$, we get for all $t\in\mathcal{G}_k$, on event $B_k\cap E$,

$$\begin{aligned}\left|\frac{1}{n}\sum_{i=1}^{n}1\{X_i\in t\}-\mathbb{P}\{X\in t\}\right| &\le 2kn^{-1}\log n+n^{-1}(\log n)^2\log\log n+2kn^{-1}\log n\\ &\le Cn^{-1}(\log n)^2\log\log n,\end{aligned}$$

which proves (59).

The proofs for (60) and (61) are essentially analogous to that for (59). For (60), by Lemma 3(i) and (ii) and the definition of $A_{k,1}$, it holds that on event $A_{k,1}\cap E$, for

all $t \in \mathcal{G}_k$ such that $\mathbb{P}\{X \in t\} > n^{-1}\log n$,

$$\begin{aligned}
&\mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\} - \mathbb{P}\{X \in t\} \right| \\
&\leq \mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\} - \frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t^{\#}\} \right| \\
&\quad + \mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t^{\#}\} - \mathbb{P}\{X \in t^{\#}\} \right| \\
&\quad + \mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \mathbb{P}\{X \in t^{\#}\} - \mathbb{P}\{X \in t\} \right| \\
&\leq n^{\frac{1}{2}}(\log n)^{-\frac{1}{2}} 2kn^{-1}\log n + n^{-\frac{1}{2}} + n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n + n^{\frac{1}{2}}(\log n)^{-\frac{1}{2}} 2k(\sup f)n^{-1-\rho} \\
&\leq Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n.
\end{aligned}$$

For (61), we have for all $t \in \mathcal{G}_k$ such that $\mathbb{P}\{X \in t\} > n^{-1}\log n$, on event $A_{k,2} \cap E$,

$$\begin{aligned}
&\mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \frac{1}{n}\sum_{i=1}^{n} Y_i 1\{X_i \in t\} - \mathbb{E}(Y1\{X \in t\}) \right| \\
&\leq \mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \frac{1}{n}\sum_{i=1}^{n} Y_i 1\{X_i \in t\} - \frac{1}{n}\sum_{i=1}^{n} Y_i 1\{X_i \in t^{\#}\} \right| \\
&\quad + \mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \frac{1}{n}\sum_{i=1}^{n} Y_i 1\{X_i \in t^{\#}\} - \mathbb{E}(Y1\{X \in t^{\#}\}) \right| \\
&\quad + \mathbb{P}\{X \in t\}^{-\frac{1}{2}} \left| \mathbb{E}(Y1\{X \in t^{\#}\}) - \mathbb{E}(Y1\{X \in t\}) \right| \\
&\leq Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n,
\end{aligned}$$

upon noticing that

$$\begin{aligned}
&\left| \frac{1}{n}\sum_{i=1}^{n} Y_i 1\{X_i \in t\} - \frac{1}{n}\sum_{i=1}^{n} Y_i 1\{X_i \in t^{\#}\} \right| \leq \frac{1}{n}\sum_{i=1}^{n} 1\{X_i \in t\Delta t^{\#}\} \quad \text{and} \\
&\left| \mathbb{E}(Y1\{X \in t^{\#}\}) - \mathbb{E}(Y1\{X \in t\}) \right| \leq \mathbb{P}\{X \in t\Delta t^{\#}\}
\end{aligned}$$

since $Y \in \{0,1\}$. This completes the proof of Lemma 5. □

Lemma 6. *Let Assumption 2, 3, and 4 hold. Let the events $A_{1,k}$, $A_{2,k}$, and E be given as in Lemma 4 and (49). Let $k = O(\log_\delta n)$ with $\delta > 1$ be given. Then, it holds that on $A_{1,k} \cap A_{2,k} \cap E$, for any $t \in \mathcal{G}_k$,*

$$\mathbb{P}\{X \in t\}^{\frac{1}{2}} \left| \overline{Y}(t) - \mathbb{E}(Y|X \in t) \right| \leq Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n$$

Proof. We have

$$
\begin{aligned}
&\left|\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right| \\
&\leq \left|\frac{\frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\}}{\frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\}} + \frac{\frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\}}{\mathbb{P}\{X \in t\}}\right| + \left|\frac{\frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\}}{\mathbb{P}\{X \in t\}} - \frac{\mathbb{E}(Y1\{X \in t\})}{\mathbb{P}\{X \in t\}}\right| \\
&= \frac{1}{\mathbb{P}\{X \in t\}}\left(\underbrace{\frac{\frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\}}{\frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\}}}_{\leq 1}\left|\mathbb{P}\{X \in t\} - \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\}\right| + \left|\frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\} - \mathbb{E}(Y1\{X \in t\})\right|\right) \\
&\leq \frac{1}{\mathbb{P}\{X \in t\}}\left(\left|\mathbb{P}\{X \in t\} - \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\}\right| + \left|\frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\} - \mathbb{E}(Y1\{X \in t\})\right|\right)
\end{aligned}
$$

and thus,

$$
\begin{aligned}
&\mathbb{P}\{X \in t\}^{\frac{1}{2}}\left|\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right| \\
&\leq \mathbb{P}\{X \in t\}^{-\frac{1}{2}}\left(\left|\mathbb{P}\{X \in t\} - \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\}\right| + \left|\frac{1}{n}\sum_{i=1}^n Y_i 1\{X_i \in t\} - \mathbb{E}(Y1\{X \in t\})\right|\right).
\end{aligned} \tag{65}
$$

For all $t \in \mathcal{G}_k$ such that $\mathbb{P}\{X \in t\} > n^{-1}\log n$, by (60) and (61) in Lemma 5, it holds that on $A_{k,1} \cap A_{k,2} \cap E$,

$$
(65) \leq Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n.
$$

On the other hand, for all $t \in \mathcal{G}_k$ such that $\mathbb{P}\{X \in t\} \leq n^{-1}\log n$, we simply have

$$
\mathbb{P}\{X \in t\}^{\frac{1}{2}}\left|\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right| \leq 2\mathbb{P}\{X \in t\}^{\frac{1}{2}} \leq 2n^{-\frac{1}{2}}(\log n)^{\frac{1}{2}}.
$$

Thus, it follows that for all $t \in \mathcal{G}_k$, on $A_{k,1} \cap A_{k,2} \cap E$,

$$
\mathbb{P}\{X \in t\}^{\frac{1}{2}}\left|\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right| \leq Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n,
$$

as was to be shown. This completes the proof. □

Lemma 7. *Let Assumption 2, 3, and 4 hold. Let the events $A_{1,k}$, $A_{2,k}$, B_k and E be given as in Lemma 4 and (49). Let $k = O(\log_\delta n)$ with $\delta > 1$ be given. For a cell t and a split s, define*

$$
\begin{aligned}
&Q'(t,s) = \mathbb{P}\{X \in t'\}\Big(\mathbb{E}(Y|X \in t') - \mathbb{E}(Y|X \in t)\Big)^2 \quad \textit{and} \\
&Q'_n(t,s) = \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t'\}\Big(\overline{Y}(t') - \overline{Y}(t)\Big)^2.
\end{aligned}
$$

$Q''(t,s)$ and $Q_n''(t,s)$ are defined analogously. Then, it holds that on $A_{1,k}\cap A_{2,k}\cap B_k\cap E$, for any $t\in\mathcal{G}_{k-1}$ and any split s of t,

$$\left|\sqrt{Q'(t,s)^{\frac{1}{2}}}-\sqrt{Q_n'(t,s)}\right|\le Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n \quad \textit{and}$$
$$\left|\sqrt{Q''(t,s)}-\sqrt{Q_n''(t,s)}\right|\le Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n.$$

Proof. By symmetry, it suffices to only prove

$$\left|\sqrt{Q'(t,s)}-\sqrt{Q_n'(t,s)}\right|\le Cn^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n \tag{66}$$

on $A_{1,k}\cap A_{2,k}\cap B_k\cap E$, for any $t\in\mathcal{G}_{k-1}$ and any split s of t. Notice that

$$\begin{aligned}
&\left|\sqrt{Q'(t,s)}-\sqrt{Q_n'(t,s)}\right| \\
&\le\left|\left(\frac{1}{n}\sum_{i=1}^n 1\{X_i\in t'\}\right)^{\frac{1}{2}}\left(\overline{Y}(t')-\overline{Y}(t)\right)-\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\left(\overline{Y}(t')-\overline{Y}(t)\right)\right| \\
&\quad+\left|\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\left(\overline{Y}(t')-\overline{Y}(t)\right)-\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\left(\mathbb{E}(Y|X\in t')-\mathbb{E}(Y|X\in t)\right)\right| \\
&\le\left|\left(\frac{1}{n}\sum_{i=1}^n 1\{X_i\in t'\}\right)^{\frac{1}{2}}-\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\right|\left|\overline{Y}(t')-\overline{Y}(t)\right| &(67)\\
&\quad+\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\left|\overline{Y}(t)-\mathbb{E}(Y|X\in t)\right| &(68)\\
&\quad+\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\left|\overline{Y}(t')-\mathbb{E}(Y|X\in t')\right| &(69)
\end{aligned}$$

For (67), we have $\left|\overline{Y}(t')-\overline{Y}(t)\right|\le 1$ and

$$\begin{aligned}
&\left|\left(\frac{1}{n}\sum_{i=1}^n 1\{X_i\in t'\}\right)^{\frac{1}{2}}-\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\right| \\
&=\left(\left(\frac{1}{n}\sum_{i=1}^n 1\{X_i\in t'\}\right)^{\frac{1}{2}}+\mathbb{P}\{X\in t'\}^{\frac{1}{2}}\right)^{-1}\left|\frac{1}{n}\sum_{i=1}^n 1\{X_i\in t'\}-\mathbb{P}\{X\in t'\}\right| \\
&\le\mathbb{P}\{X\in t'\}^{-\frac{1}{2}}\left|\frac{1}{n}\sum_{i=1}^n 1\{X_i\in t'\}-\mathbb{P}\{X\in t'\}\right|.
\end{aligned}$$

Notice that if $t\in\mathcal{G}_{k-1}$, then $t'\in\mathcal{G}_k$. For all $t'\in\mathcal{G}_k$ such that $\mathbb{P}\{X\in t'\}>n^{-1}\log n$,

by (60) of Lemma 5, we have on event $A_{k,1} \cap E$ that

$$\begin{aligned}
&\left| \left(\frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t'\} \right)^{\frac{1}{2}} - \mathbb{P}\{X \in t'\}^{\frac{1}{2}} \right| \\
&= \left(\left(\frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t'\} \right)^{\frac{1}{2}} + \mathbb{P}\{X \in t'\}^{\frac{1}{2}} \right)^{-1} \left| \frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t'\} - \mathbb{P}\{X \in t'\} \right| \\
&\leq \mathbb{P}\{X \in t'\}^{-\frac{1}{2}} \left| \frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t'\} - \mathbb{P}\{X \in t'\} \right| \\
&\leq C n^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} \log\log n,
\end{aligned}$$

and thus (67) $\leq C n^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} \log\log n$ since $\left| \overline{Y}(t') - \overline{Y}(t) \right| \leq 1$.

On the other hand, for all $t' \in \mathcal{G}_k$ such that $\mathbb{P}\{X \in t'\} \leq n^{-1} \log n$, by (59) of Lemma 5, on event $B_k \cap E$,

$$\left| \frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t'\} - \mathbb{P}\{X \in t'\} \right| \leq C n^{-1} (\log n)^2 \log\log n.$$

Then,

$$\begin{aligned}
\frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t'\} &\leq \mathbb{P}\{X \in t'\} + C n^{-1} (\log n)^2 \log\log n \\
&\leq C n^{-1} (\log n)^2 \log\log n,
\end{aligned}$$

from which it follows that

$$\begin{aligned}
(67) &\leq \left(\frac{1}{n} \sum_{i=1}^{n} 1\{X_i \in t'\} \right)^{\frac{1}{2}} + \mathbb{P}\{X \in t'\}^{\frac{1}{2}} \\
&\leq C n^{-\frac{1}{2}} (\log n)(\log\log n)^{\frac{1}{2}}
\end{aligned}$$

for all $t' \in \mathcal{G}_k$ such that $\mathbb{P}\{X \in t'\} \leq n^{-1} \log n$. Therefore, (67) $\leq C n^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} \log\log n$ for all $t \in \mathcal{G}_{k-1}$ and all split s of t.

Next, for (68), notice that $\mathbb{P}\{X \in t'\} \leq \mathbb{P}\{X \in t\}$ and since $t' \subset t$. Also, if $t \in \mathcal{G}_{k-1}$, then $t \in \mathcal{G}_k$ since $\mathcal{G}_i \subset \mathcal{G}_j$ for $i \leq j$. Hence, by Lemma 6, on $A_{k,1} \cap A_{k,2} \cap E$, for all $t \in \mathcal{G}_{k-1}$, we have

$$(68) \leq C n^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} \log\log n.$$

Similarly, we have (69) $\leq C n^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} \log\log n$ on $A_{k,1} \cap A_{k,2} \cap E$, for all $t \in \mathcal{G}_{k-1}$, from which (66) follows. This completes the proof. □

Lemma 8. *Let Assumption 2, 3, and 4 hold. Let the events $A_{1,k}$, $A_{2,k}$, B_k and E be given as in Lemma 4 and (49). Let $k = O(\log_\delta n)$ with $\delta > 1$ be given. Then, it holds that on $A_{1,k} \cap A_{2,k} \cap B_k \cap E$, for any $\epsilon > 0$ and any $t \in \mathcal{G}_{k-1}$,*

$$Q(t, s_n) \geq \frac{1}{(1+\epsilon)^2} Q(t, s^*) - C\frac{1}{\epsilon} n^{-1} (\log n)^3 (\log\log n)^2.$$

Proof. First, we will prove that on $A_{1,k} \cap A_{2,k} \cap B_k \cap E$, for any $\epsilon > 0$, any $t \in \mathcal{G}_{k-1}$ and any split s of t,

$$Q_n(t, s) \geq \frac{1}{1+\epsilon} Q(t, s) - C\frac{1}{\epsilon} n^{-1} (\log n)^3 (\log\log n)^2 \quad \text{and} \tag{70}$$

$$Q(t, s) \geq \frac{1}{1+\epsilon} Q_n(t, s) - C\frac{1}{\epsilon} n^{-1} (\log n)^3 (\log\log n)^2. \tag{71}$$

By symmetry and some rearrangement of the terms, it suffices to show that on $A_{1,k} \cap A_{2,k} \cap B_k \cap E$, for any $\epsilon > 0$, for any $t \in \mathcal{G}_{k-1}$ and any split s of t,

$$Q'(t, s) \leq (1+\epsilon) Q'_n(t, s) + C\left(1 + \frac{1}{\epsilon}\right) n^{-1} (\log n)^3 (\log\log n)^2 \quad \text{and} \tag{72}$$

$$Q'_n(t, s) \leq (1+\epsilon) Q'(t, s) + C\left(1 + \frac{1}{\epsilon}\right) n^{-1} (\log n)^3 (\log\log n)^2, \tag{73}$$

where $Q'(t, s)$ and $Q'_n(t, s)$ are defined in Lemma 7.

By Lemma 7, on $A_{1,k} \cap A_{2,k} \cap B_k \cap E$, for any $\epsilon > 0$, for any $t \in \mathcal{G}_{k-1}$ and any split s of t,

$$\begin{aligned}
Q'(t, s) &\leq \left(\sqrt{Q'_n(t, s)} + Cn^{-\frac{1}{2}} (\log n)^{\frac{3}{2}} \log\log n\right)^2 \\
&= Q'_n(t, s) + Cn^{-1} (\log n)^3 (\log\log n)^2 + \sqrt{Q'_n(t, s)} Cn^{-\frac{1}{2}} (\log)^{\frac{3}{2}} \log\log n \\
&\leq (1+\epsilon) Q'_n(t, s) + C\left(1 + \frac{1}{\epsilon}\right) n^{-1} (\log)^3 (\log\log n)^2,
\end{aligned}$$

where the second inequality follows from

$$\begin{aligned}
&\sqrt{Q'_n(t, s)} Cn^{-\frac{1}{2}} (\log)^{\frac{3}{2}} \log\log n \\
&= 2\left(\epsilon^{\frac{1}{2}} \sqrt{Q'_n(t, s)}\right)\left(\epsilon^{-\frac{1}{2}} Cn^{-\frac{1}{2}} (\log)^{\frac{3}{2}} \log\log n\right) \\
&\leq \epsilon Q'_n(t, s) + \frac{1}{\epsilon} Cn^{-1} (\log n)^3 (\log\log n)^2
\end{aligned}$$

for every $\epsilon > 0$, thus proving (72) and analogously, (73). This proves (70) and (71).

Then, we can deduced that

$$\begin{aligned}
Q(t,s_n) &\geq \frac{1}{1+\epsilon}Q_n(t,s_n) - C\frac{1}{1+\epsilon}\left(1+\frac{1}{\epsilon}\right)n^{-1}(\log n)^3(\log\log n)^2 \\
&\geq \frac{1}{1+\epsilon}Q_n(t,s^*) - C\frac{1}{\epsilon}n^{-1}(\log n)^3(\log\log n)^2 \\
&\geq \frac{1}{(1+\epsilon)^2}Q(t,s^*) - C\frac{1}{1+\epsilon}\frac{1}{\epsilon}n^{-1}(\log n)^3(\log\log n)^2 - C\frac{1}{\epsilon}n^{-1}(\log n)^3(\log\log n)^2 \\
&\geq \frac{1}{(1+\epsilon)^2}Q(t,s^*) - C\frac{1}{\epsilon}n^{-1}(\log n)^3(\log\log n)^2,
\end{aligned}$$

where the first inequality follows from (71), the second inequality is due to the fact that $Q_n(t,s_n) \geq Q_n(t,s^*)$ by the definition of s_n, the second to last inequality holds by (70), and the last inequality holds since $\epsilon > 0$ so that $\frac{1}{1+\epsilon}\frac{1}{\epsilon} < \frac{1}{\epsilon}$. This completes the proof. □

Lemma 9. *Let $k = \lfloor \eta \log_\delta n \rfloor$ with $\eta > 0$ and $\delta > 1$. Let $\lambda, \gamma \in (0,1]$ and $\epsilon = (\log n)^{-1}$. Then, we have*

$$\left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^k = \begin{cases} O\left(n^{-\eta\log_\delta\left(\frac{1}{1-\gamma\lambda}\right)}\right) & \text{if } \gamma\lambda \neq 1 \\ o(n^{-C}) & \text{if } \gamma\lambda = 1 \end{cases}$$

for every $C > 0$.

Proof. First, consider the case when $\gamma\lambda \neq 1$. If we can show that

$$\left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^k \leq C(1-\gamma\lambda)^k,$$

then it follows from[28]

$$(1-\gamma\lambda)^k \sim (1-\gamma\lambda)^{\log_\delta n^\eta} = n^{-\eta\log_\delta\left(\frac{1}{1-\gamma\lambda}\right)}$$

that

$$\left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^k = O\left(n^{-\eta\log_\delta\left(\frac{1}{1-\gamma\lambda}\right)}\right).$$

Straightforward calculation yields

$$\frac{1 - \frac{\gamma\lambda}{(1+\epsilon)^2}}{1-\gamma\lambda} = 1 + \frac{\gamma\lambda}{1-\gamma\lambda}\frac{1+\frac{2}{\epsilon}}{1+\frac{2}{\epsilon}+\frac{1}{\epsilon^2}}.$$

[28] Here we use the equality that $a^{\log_\delta b} = b^{\log_\delta a}$ for all $a, b, \delta > 0$ and $\delta \neq 1$.

Then, setting $\epsilon = (\log n)^{-1}$, we get

$$\log \frac{\left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^k}{(1-\gamma\lambda)^k} = k \log \left(1 + \frac{\gamma\lambda}{1-\gamma\lambda} \frac{1 + 2\log n}{1 + 2\log n + (\log n)^2}\right)$$
$$\sim 2k \frac{\gamma\lambda}{1-\gamma\lambda} (\log n)^{-1} \sim \frac{2\eta\gamma\lambda}{1-\gamma\lambda} \frac{1}{\log \delta},$$

where we have used the fact that $\log(1+x) \sim x$ as $x \to 0$. This gives

$$\left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^k \sim \exp\left(\frac{2\eta\gamma\lambda}{1-\gamma\lambda} \frac{1}{\log \delta}\right) (1-\gamma\lambda)^k,$$

as was to be shown.

Next, we consider the case when $\gamma\lambda = 1$, or equivalently, $\gamma = \lambda = 1$. Then,

$$\left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right) = \left(\frac{1 + 2\log n}{1 + 2\log n + (\log n)^2}\right)^k$$
$$\sim \left(\frac{2}{\log n}\right)^{\log_\delta n^\eta} = n^{\eta \log_\delta 2 - \eta \log_\delta \log n} = o(n^{-C})$$

for every $C > 0$, i.e., $n^{\eta \log_\delta 2 - \eta \log_\delta \log n}$ decays faster than any polynomial order. This completes the proof. □

C Proofs of Main Results

C.1 Proof of Theorem 1

Proof. Let $E_1 = A_{k,1} \cap A_{k,2} \cap E$, where the events are given in Lemma 3 and (49). Then,

$$\mathbb{E}\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2$$
$$= \mathbb{E}\left(\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2 1(E_1)\right) + \mathbb{E}\left(\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2 1(E_1^c)\right) \tag{74}$$

By Lemma 3(iii) and Lemma 4, we can bound the second term in (74) by

$$\mathbb{E}\left(\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2 1(E_1^c)\right)$$
$$\leq \mathbb{P}(E_1^c) \leq \mathbb{P}(A_{k,1}^c) + \mathbb{P}(A_{k,2}^c) + \mathbb{P}(E^c)$$
$$\leq 6 \exp\left(-C(\log n)^2 \log\log n\right) + \max(1, \sup f) d_1 \lceil n^{1+\rho} \rceil \left(\frac{\sup f}{n^\rho}\right)^{\log n} \tag{75}$$

for a given $\rho > 0$. By Assumption 3, $d_1 = O(n^{C_1})$ for some $C_1 > 0$, and thus, it is easy to see that (75) vanishes faster than any polynomial rate in n, i.e., (75) $= o(n^{-C})$ for any $C > 0$.

It remains to show the first item in (74) $= O(n^{-1+\eta \log_\delta 2}(\log n)^3(\log\log n)^2)$. By the definition of $\widehat{p}_{\widehat{T}}(X)$ and $p^*_{\widehat{T}}(X)$, we can deduce that

$$\begin{aligned}
&\mathbb{E}\left(\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2 1(E_1)\right) \\
&= \mathbb{E}\left[\mathbb{E}\left(\left(\widehat{p}_{\widehat{T}}(X) - p^*_{\widehat{T}}(X)\right)^2 1(E_1)\middle| \mathcal{D}, \Theta\right)\right] \\
&= \mathbb{E}\left[\sum_{t\in\widehat{T}} \mathbb{E}\left(1\{X \in t\}\left(\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right)^2 1(E_1)\middle| \mathcal{D}, \Theta\right)\right], \qquad (76)
\end{aligned}$$

where we have suppressed Θ from $\widehat{T}(\Theta)$ for simplicity. Notice that given specific $\mathcal{D}$ and Θ, $\overline{Y}(t) - \mathbb{E}(Y|X \in t)$ and $1(E_1)$ becomes nonrandom, while $1\{X \in t\}$ remains random with respect to the σ-field generated by X. Therefore, it follows that

$$\begin{aligned}
(76) &= \mathbb{E}\left(\sum_{t\in\widehat{T}}\left(\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right)^2 1(E_1)\mathbb{E}(1\{X \in t\}|\mathcal{D}, \Theta)\right) \\
&= \mathbb{E}\left(\sum_{t\in\widehat{T}}\left(\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right)^2 1(E_1)\mathbb{P}\{X \in t\}\right) \\
&\le Cn^{-1}(\log n)^3(\log\log n)^2\mathbb{E}\left(\sum_{t\in\widehat{T}} 1(E_1)\right) \\
&\le C2^k n^{-1}(\log n)^3(\log\log n)^2 \\
&= O(n^{-1+\eta\log_\delta 2}(\log n)^3(\log\log n)^2),
\end{aligned}$$

where the first inequality is due to Lemma 6 and the second inequality holds since there are at most 2^k leaves for a tree of depth k. This completes the proof. □

C.2 Proof of Theorem 2

Proof. Define $E_2 = A_{k,1} \cap A_{k,2} \cap B_k \cap E$. Then,

$$\begin{aligned}
&\mathbb{E}\left(p^*_{\widehat{T}}(X) - p(X)\right)^2 \\
&= \mathbb{E}\left(\left(p^*_{\widehat{T}}(X) - p(X)\right)^2 1(E_2)\right) + \mathbb{E}\left(\left(p^*_{\widehat{T}}(X) - p(X)\right)^2 1(E_2^c)\right) \qquad (77)
\end{aligned}$$

Similar to (75) in the proof of Theorem 1, the second item in (77) is $o(n^{-C})$ for any $C > 0$. Therefore, it remains to derive the stated rates for the first item of (77).

We first note that

$$
\begin{aligned}
&\mathbb{E}\left(\left(p^*_{\widehat{T}}(X)-p(X)\right)^2 1(E_2)\right)\\
&=\sum_{\theta^{1:k}}\mathbb{P}\{\Theta^{1:k}=\theta^{1:k}\}\mathbb{E}\left(\left(p^*_{\widehat{T}}(X)-p(X)\right)^2 1(E_2)\middle|\Theta^{1:k}=\theta^{1:k}\right)\\
&=\sum_{\theta^{1:k}}\mathbb{P}\{\Theta^{1:k}=\theta^{1:k}\}\mathbb{E}\left[\mathbb{E}\left(\sum_{t\in\widehat{T}(\theta^{1:k})}1\{X\in t\}\left(\mathbb{E}(Y|X\in t)-p(X)\right)^2 1(E_2)\middle|\mathcal{D},\Theta^{1:k}=\theta^{1:k}\right)\right]\\
&=\sum_{\theta^{1:k}}\mathbb{P}\{\Theta^{1:k}=\theta^{1:k}\}\mathbb{E}\left[\sum_{t\in\widehat{T}(\theta^{1:k})}\mathbb{P}\left(X\in t\middle|\mathcal{D},\Theta^{1:k}=\theta^{1:k}\right)\mathrm{var}\left(p(X)\middle|X\in t,\mathcal{D},\Theta^{1:k}=\theta^{1:k}\right)1(E_2)\right],
\end{aligned}
\tag{78}
$$

where the first equality is due to the law of total expectation and the third equality follows from the fact that $1(E_2)$ is nonrandom given $\mathcal{D}$ and $\Theta^{1:k}$ and

$$
\begin{aligned}
&\mathbb{E}\left(1\{X\in t\}\left(\mathbb{E}(Y|X\in t)-p(X)\right)^2\middle|\mathcal{D},\Theta^{1:k}\right)\\
&=\mathbb{E}\left(1\{X\in t\}\left(\mathbb{E}(Y|X\in t,\mathcal{D},\Theta^{1:k})-p(X)\right)^2\middle|\mathcal{D},\Theta^{1:k}\right)\\
&=\mathbb{P}(X\in t|\mathcal{D},\Theta^{1:k})\mathbb{E}\left[\left(\mathbb{E}(p(X)|X\in t,\mathcal{D},\Theta^{1:k})-p(X)\right)^2\middle|X\in t,\mathcal{D},\Theta^{1:k}\right]\\
&=\mathbb{P}(X\in t|\mathcal{D},\Theta^{1:k})\mathrm{var}\left(p(X)\middle|X\in t,\mathcal{D},\Theta^{1:k}\right)
\end{aligned}
$$

In the sense of Remark 2, for simplicity, we write (78) as

$$
(78)=\mathbb{E}\left[\sum_{\theta^{1:k}}\mathbb{P}\{\Theta^{1:k}=\theta^{1:k}\}\sum_{t\in\widehat{T}(\theta^{1:k})}\mathbb{P}\{X\in t\}\mathrm{var}(p(X)|X\in t)1(E_2)\right].
$$

Denote by Θ^ℓ_t the column subsampling of the cell t at level ℓ that splits into the child cells at level $\ell+1$ and θ^ℓ_t its realization. Then, on event E_2, it holds that

$$
\begin{aligned}
&\sum_{\theta^{1:k}}\mathbb{P}\{\Theta^{1:k}=\theta^{1:k}\}\sum_{t\in\widehat{T}(\theta^{1:k})}\mathbb{P}\{X\in t\}\mathrm{var}(p(X)|X\in t)\\
&=\sum_{\theta^{1:k}}\mathbb{P}\{\Theta^{1:k}=\theta^{1:k}\}\sum_{t\in\widehat{T}(\theta^{1:k-1})}\left(\mathbb{P}\{X\in t'\}\mathrm{var}(p(X)|X\in t')+\mathbb{P}\{X\in t''\}\mathrm{var}(p(X)|X\in t'')\right)\\
&=\sum_{\theta^{1:k-1}}\mathbb{P}\{\Theta^{1:k-1}=\theta^{1:k-1}\}\sum_{t\in\widehat{T}(\theta^{1:k-1})}\sum_{\theta^k_t}\mathbb{P}\{\Theta^k_t=\theta^k_t\}\left(\mathbb{P}\{X\in t\}\mathrm{var}(p(X)|X\in t)-Q(t,s_n)\right)
\end{aligned}
\tag{79}
$$

where the first equality is just a rearrangement of the summands, and the second inequality is due to Lemma 1. By Lemma 8, we have on event E_2, for any given

$\epsilon > 0$,

(79)

$$\leq \sum_{\theta^{1:k-1}} \mathbb{P}\{\Theta^{1:k-1} = \theta^{1:k-1}\} \sum_{t \in \widehat{T}(\theta^{1:k-1})} \sum_{\theta_t^k} \mathbb{P}\{\Theta_t^k = \theta_t^k\}$$
$$\times \left[\mathbb{P}\{X \in t\} \mathrm{var}(p(X)|X \in t) - \frac{1}{(1+\epsilon)^2} Q(t, s^*) + C \frac{1}{\epsilon} n^{-1} (\log n)^3 (\log\log n)^2 \right]$$
$$\leq \sum_{\theta^{1:k-1}} \mathbb{P}\{\Theta^{1:k-1} = \theta^{1:k-1}\} \sum_{t \in \widehat{T}(\theta^{1:k-1})} \sum_{\theta_t^k} \mathbb{P}\{\Theta_t^k = \theta_t^k\} \left[\mathbb{P}\{X \in t\} \mathrm{var}(p(X)|X \in t) - \frac{1}{(1+\epsilon)^2} Q(t, s^*) \right]$$
$$+ C 2^{k-1} \frac{1}{\epsilon} n^{-1} (\log n)^3 (\log\log n)^2, \tag{80}$$

where the second inequality holds since there are at most 2^{k-1} cells in $\widehat{T}(\theta^{1:k-1})$.

Recall that s^* is chosen that maximizes $Q(t, s)$ with column restrictions, while s_0 is chosen without column restrictions. For a cell t at an arbitrary level, let Θ_t be the column subsampling of t that splits into the child cells in the next level and let θ_t be the realization of Θ_t. If at least one of the optimal coordinates is included in Θ_t, then $Q(t, s^*) = Q(t, s_0)$[29] and thus, by Assumption 1,

$$\begin{aligned} &\mathbb{P}\{X \in t\} \mathrm{var}(p(X)|X \in t) - \frac{1}{(1+\epsilon)^2} Q(t, s^*) \\ &\leq \left(1 - \frac{\lambda}{(1+\epsilon)^2}\right) \mathbb{P}\{X \in t\} \mathrm{var}(p(X)|X \in t). \end{aligned} \tag{81}$$

On the other hand, if none of the optimal coordinates are included in Θ_t, we simply have

$$\mathbb{P}\{X \in t\} \mathrm{var}(p(X)|X \in t) - \frac{1}{(1+\epsilon)^2} Q(t, s^*) \leq \mathbb{P}\{X \in t\} \mathrm{var}(p(X)|X \in t). \tag{82}$$

since $Q(t, s) \geq 0$ for all s.

By the definition of column subsampling, it is not difficult to see that for every cell t, the probability that Θ_t contains at least one of the optimal coordinates is at least γ. Therefore, we may define an event $F_t = \{\theta_t \subset \{1, \cdots, d\} : Q(t, s^*) = Q(t, s_0)\}$.

[29] Due to the rule of breaking ties, s^* needs not equal s_0 in this case.

On F_t, the split s^* of t yields the same variance reduction as s_0. Then,

$$\begin{aligned}
&\sum_{\theta_t^k} \mathbb{P}\{\Theta_t^k = \theta_t^k\} \left[\mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) - \frac{1}{(1+\epsilon)^2} Q(t, s^*)\right] \\
&= \sum_{\theta_t^k \in F_t} \mathbb{P}\{\Theta_t^k = \theta_t^k\} \left[\mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) - \frac{1}{(1+\epsilon)^2} Q(t, s^*)\right] \\
&\quad + \sum_{\theta^k \notin F_t} \mathbb{P}\{\Theta_t^k = \theta_t^k\} \left[\mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) - \frac{1}{(1+\epsilon)^2} Q(t, s^*)\right] \\
&\leq \sum_{\theta_t^k \in F_t} \mathbb{P}\{\Theta_t^k = \theta_t^k\} \left(1 - \frac{\lambda}{(1+\epsilon)^2}\right) \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) \\
&\quad + \sum_{\theta^k \notin F_t} \mathbb{P}\{\Theta_t^k = \theta_t^k\} \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t) \\
&\leq \left(1 - \frac{\nu\lambda}{(1+\epsilon)^2}\right) \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t), \qquad (83)
\end{aligned}$$

where $\nu = \sum_{\theta_t^k \in F_t} \mathbb{P}\{\Theta_t^k = \theta_t^k\}$, the first inequality is due to (81) and (82), and the second inequality holds since $\sum_{\theta^k \notin F_t} \mathbb{P}\{\Theta_t^k = \theta_t^k\} = 1 - \nu$ and $\mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t)$ only depends on θ^{k-1}, not θ^k. Since (83) is monotone decreasing in ν and $\nu \geq \gamma$, we get

$$(83) \leq \left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right) \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t). \qquad (84)$$

Combined with (84), (80) can be further bounded as

$$\begin{aligned}
&(80) \\
&\leq \left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right) \sum_{\theta^{1:k-1}} \mathbb{P}\{\Theta^{1:k-1} = \theta^{1:k-1}\} \sum_{t \in \widehat{T}(\theta^{1:k-1})} \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t), \\
&\leq \left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^{k-1} \sum_{\theta^{1:1}} \mathbb{P}\{\Theta^{1:1} = \theta^{1:1}\} \sum_{t \in \widehat{T}(\theta^{1:1})} \mathbb{P}\{X \in t\}\mathrm{var}(p(X)|X \in t), \\
&\quad + C(2 + \cdots + 2^{k-2} + 2^{k-1}) \frac{1}{\epsilon} n^{-1} (\log n)^3 (\log\log n)^2 \\
&\leq \left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^{k} \mathbb{P}\{X \in [0,1]^d\}\mathrm{var}(p(X)|X \in [0,1]^d) + C(2^k - 1) \frac{1}{\epsilon} n^{-1} (\log n)^3 (\log\log n)^2 \\
&\leq \mathrm{var}(p(X)) \left(1 - \frac{\gamma\lambda}{(1+\epsilon)^2}\right)^{k} + C \frac{1}{\epsilon} n^{-1+\eta \log_\delta 2} (\log n)^3 (\log\log n)^2, \qquad (85)
\end{aligned}$$

where the second and the third inqualities follows from a recursive application of the steps in (79) to (84) upward until level zero and the last inequality holds since $X \in [0,1]^d$ with probability one and $k \leq \lfloor \eta \log_\delta 2 \rfloor$ so that $2^k \leq n^{\eta \log_\delta 2}$.

By Lemma 9, setting $\epsilon = (\log n)^{-1}$, we have

$$(85) = \begin{cases} O\left(n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)} + n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2\right) & \text{if } \gamma\lambda \neq 1 \\ O(n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2) & \text{if } \gamma\lambda = 1 \end{cases},$$

as was to be shown. This completes the proof of Theorem 2. □

C.3 Proof of Theorem 3

Proof. Define

$$B_{k,2} = \bigcap_{t \in \mathcal{G}_k^\#} \left\{ \left| \frac{1}{n} \sum_{i=1}^n 1\{X_i \in t\} - \mathbb{P}\{X \in t\} \right| \leq n^{-\frac{1}{2}} (\log n) \log\log n \right\}.$$

Then, by Hoeffding's inequality, we have for every $\epsilon > 0$ that

$$\mathbb{P}\left\{ \left| \frac{1}{n} \sum_{i=1}^n 1\{X_i \in t\} - \mathbb{P}\{X \in t\} \right| \geq \epsilon \right\} \leq 2\exp\left(-2n\epsilon^2\right).$$

Setting $\epsilon = n^{-\frac{1}{2}} (\log n) \log\log n$, we get

$$\mathbb{P}\left\{ \left| \frac{1}{n} \sum_{i=1}^n 1\{X_i \in t\} - \mathbb{P}\{X \in t\} \right| \geq n^{-\frac{1}{2}} (\log n) \log\log n \right\} \leq 2\exp\left(-2(\log n)^2(\log\log n)^2\right).$$

Using the arguments in (57) in the proof of Lemma 4, we can conclude that

$$\begin{aligned} \mathbb{P}(B_{k,2}^c) &\leq \exp(C(\log n)^2) \times 2\exp\left(-2(\log n)^2(\log\log n)^2\right) \\ &\leq 2\exp(-C(\log n)^2(\log\log n)^2), \end{aligned}$$

which decays faster than any polynomial rate.

Now, on event $A_{k,1} \cap A_{k,2} \cap B_k \cap B_{k,2} \cap E$, we claim that for every $t \in \widehat{T}$,

$$\frac{\underline{C}}{2} \leq \overline{Y}(t) \leq \frac{1+\overline{C}}{2} \tag{86}$$

for all large enough n, where $\overline{C}$ and $\underline{C}$ are given in Assumption 5. We defer the proof of (86) to the end of the proof of Theorem 3.

Let $E_3 = A_{k,1} \cap A_{k,2} \cap B_k \cap B_{k,2} \cap E$. Then, on E_3, we have for each $t \in \widehat{T}$ that

$$\begin{aligned}
&\mathbb{E}\left[\left(\widehat{g}_{\widehat{T}}(X) - g(X)\right)^2 1(E_3)\right] \\
&= \mathbb{E}\left[\sum_{t \in \widehat{T}} 1\{X \in t\}\left(F^{-1}(\overline{Y}(t)) - F^{-1}(p(X))\right)^2 1(E_3)\right] \\
&\leq C\mathbb{E}\left(\sum_{t \in \widehat{T}} 1\{X \in t\}\left(\overline{Y}(t) - p(X)\right)^2 1(E_3)\right) \\
&= C\mathbb{E}\left[\left(\widehat{p}_{\widehat{T}}(X) - p(X)\right)^2 1(E_3)\right]
\end{aligned}$$

where the first inequality holds since F^{-1} is Lipschitz continuous on $[\underline{C}/2, (1+\overline{C})/2]$. Therefore, the convergence rate for the $\widehat{g}_{\widehat{T}}$ is the same as that for the conditional choice probability $\widehat{p}_{\widehat{T}}$.

On the other hand, on E_3^c, similar to the arguments in (75),

$$\mathbb{E}\left[\left(\widehat{g}_{\widehat{T}}(X) - g(X)\right)^2 1(E_3^c)\right] = o(n^{-C})$$

decays faster than any polynomial order. Then, by Corollary 1, the convergence rate for $\widehat{g}_{\widehat{T}}$ is

$$\mathbb{E}\left(\widehat{g}_{\widehat{T}}(X) - g(X)\right)^2 = \begin{cases} O\left(n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)} + n^{-1+\eta\log_\delta 2}(\log n)^4(\log\log n)^2\right) & \text{if } \gamma\lambda \neq 1 \\ O(n^{-1+\eta\log_\delta 2}(\log n)^4(\log\log n)^2) & \text{if } \gamma\lambda = 1 \end{cases}.$$

It remain to prove the claim in (86). On event E, it holds that for every $t \in \widehat{T}$,

$$\begin{aligned}
\left|\frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\} - \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t^\#\}\right| &\leq \frac{1}{n}\sum_{i=1}^n 1\{X \in t\Delta t^\#\} \\
&\leq 2kn^{-1}(\log n) \leq Cn^{-1}(\log n)^2,
\end{aligned}$$

where the second inequality is due to Lemma 3(ii) and the last inequality holds since $k \leq \eta \log_\delta n$. Then, under Assumption 7, we can deduce that for all large enough n,

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^n 1\{X_i \in t^\#\} &\geq \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t\} - Cn^{-1}(\log n)^2 \\
&\geq \frac{1}{2}n^{-\alpha}.
\end{aligned}$$

Analogously, we have on $B_{k,2}$ that for every $t \in \widehat{T}$,

$$\begin{aligned}
\mathbb{P}\{X \in t^\#\} &> \frac{1}{n}\sum_{i=1}^n 1\{X_i \in t^\#\} - Cn^{-\frac{1}{2}}(\log n)\log\log n \\
&\geq \frac{1}{2}n^{-\alpha} - Cn^{-\frac{1}{2}}(\log n)\log\log n \geq \frac{1}{3}n^{-\alpha}
\end{aligned}$$

all large enough n. Upon noticing that by Lemma 3(i),

$$\left|\mathbb{P}\{X \in t\} - \mathbb{P}\{X \in t^{\#}\}\right| \leq \mathbb{P}\{X \in t\Delta t^{\#}\} \leq Cn^{-1}(\log n),$$

we can deduce that for every $t \in \widehat{T}$,

$$\mathbb{P}\{X \in t\} \geq \frac{1}{4}n^{-\alpha}$$

for all large enough n. Therefore, by Lemma 6, for every $t \in \widehat{T}$,

$$\begin{aligned}\left|\overline{Y}(t) - \mathbb{E}(Y|X \in t)\right| &\leq C(\mathbb{P}\{X \in t\})^{-\frac{1}{2}}n^{-\frac{1}{2}}(\log n)^{\frac{3}{2}}\log\log n \\ &\leq Cn^{\frac{\alpha-1}{2}}(\log n)^{\frac{3}{2}}\log\log n\end{aligned}$$

for all large enough n. By Assumption 5, we get for every $t \in \widehat{T}$,

$$\begin{aligned}\overline{Y}(t) &\leq \mathbb{E}(Y|X \in t) + Cn^{\frac{\alpha-1}{2}}(\log n)^{\frac{3}{2}}\log\log n \\ &\leq \overline{C} + Cn^{\frac{\alpha-1}{2}}(\log n)^{\frac{3}{2}}\log\log n \leq \frac{1+\overline{C}}{2} < 1\end{aligned}$$

and

$$\begin{aligned}\overline{Y}(t) &\geq \mathbb{E}(Y|X \in t) + Cn^{\frac{\alpha-1}{2}}(\log n)^{\frac{3}{2}}\log\log n \\ &\geq \underline{C} + Cn^{\frac{\alpha-1}{2}}(\log n)^{\frac{3}{2}}\log\log n \geq \frac{\underline{C}}{2} > 0\end{aligned}$$

for all large enough n. This proves (86) and completes the proof of Theorem 3. □

C.4 Proof of Theorem 4

Proof. It can be deduced that

$$\begin{aligned}\mathbb{E}\left(g(X) - \widehat{g}(X)\right)^2 &= \mathbb{E}\left(g(X) - \frac{1}{A}\sum_{a=1}^{A}\widehat{g}_{\widehat{T}_a}(X)\right)^2 \\ &\leq \frac{1}{A}\sum_{a=1}^{A}\mathbb{E}\left(g(X) - \widehat{g}_{\widehat{T}_a}(X)\right)^2,\end{aligned}$$

where $\widehat{g}_{\widehat{T}_a}, a = 1, \cdots, A$ denotes the individual tree estimate and the inequality is due to Jensen's inequality. Then, it suffices to show that each of the tree estimator has the desired rate of convergence, which follows immediately from Theorem 3.

If in addition, $\lambda \in (0, 1)$, then the optimal rate is obtained by letting η be such that

$$-\eta\log_\delta\frac{1}{1-\gamma\lambda} = -1 + \eta\log_\delta 2.$$

A straightforward calculation yields $\eta = \left(\log_\delta \frac{2}{1-\gamma\lambda}\right)^{-1}$ and thus,

$$\eta \log_\delta \frac{1}{1-\gamma\lambda} = \frac{\log_\delta \frac{1}{1-\gamma\lambda}}{\log_\delta \frac{2}{1-\gamma\lambda}} = 1 - \frac{\frac{\log 2}{\log \delta}}{\frac{\log \frac{2}{1-\gamma\lambda}}{\log \delta}}$$
$$= 1 - \left(\log_2 \frac{2}{1-\gamma\lambda}\right)^{-1} = 1 - \frac{1}{1-\log_2(1-\gamma\lambda)},$$

which takes value in $(0,1)$ and does not depend on δ. Therefore, we have

$$\mathbb{E}\left(g(X) - \widehat{g}_{\widehat{T}_a}(X)\right)^2 = O\left(n^{-\eta \log_\delta\left(\frac{1}{1-\gamma\lambda}\right)} + n^{-1+\eta \log_\delta 2}(\log n)^4(\log\log n)^2\right)$$
$$= O\left(n^{-1+\frac{1}{1-\log_2(1-\gamma\lambda)}}(\log n)^4(\log\log n)^2\right),$$

as was to be shown. This completes the proof. □

D Additional Tables and Figures

Table 5: Summary statistics: LendingClub sample

Variable / category	n	Mean / share	SD	Min.	Max.
Panel A: Outcome and numerical covariates					
Default indicator	1,213,853	0.198	0.398	0	1
FICO score (lower bound)	1,213,853	695.405	31.271	660	845
Revolving utilization (%)	1,213,853	52.128	24.308	0.000	892.300
Revolving balance ($1,000)	1,213,853	16.041	19.518	0.000	959.754
Delinquencies in past 2 years	1,213,853	0.326	0.891	0	39
Open credit lines	1,213,853	11.727	5.490	1	90
Derogatory public records	1,213,853	0.214	0.602	0	86
Loan amount ($1,000)	1,213,853	14.589	8.677	1.000	40.000
Loan term (months)	1,213,853	41.901	10.334	36	60
Annual income ($1,000)	1,213,853	74.717	38.898	0.600	260.000
Debt-to-income ratio (%)	1,213,853	18.280	8.448	0.000	60.000
Credit inquiries in past 6 months	1,213,853	0.649	0.933	0	8
Origination year	1,213,853	2015.092	1.438	2012	2018
Credit history (years)	1,213,853	15.982	7.183	2.998	83.250
Employment length (coded years)	1,213,853	5.995	3.691	0	10

Continued on next page

Summary statistics (continued)

Variable / category	n	Mean / share	SD	Min.	Max.
Panel B: Categorical covariates					
Income verification status					
Not verified	372,170	30.66%	–	–	–
Source verified	488,874	40.27%	–	–	–
Verified	352,809	29.07%	–	–	–
Home ownership					
Mortgage	602,736	49.65%	–	–	–
Rent	486,335	40.07%	–	–	–
Own	124,782	10.28%	–	–	–
Initial listing status					
f	486,179	40.05%	–	–	–
w	727,674	59.95%	–	–	–
Application type					
Individual	1,192,997	98.28%	–	–	–
Joint application	20,856	1.72%	–	–	–
Loan purpose					
Car	12,158	1.00%	–	–	–
Credit card	269,375	22.19%	–	–	–
Debt consolidation	712,445	58.69%	–	–	–
Education	1	$< 0.01\%$	–	–	–
Home improvement	76,286	6.28%	–	–	–
House	6,322	0.52%	–	–	–
Major purchase	25,303	2.08%	–	–	–
Medical	13,410	1.10%	–	–	–
Moving	8,226	0.68%	–	–	–
Other	67,441	5.56%	–	–	–
Renewable energy	776	0.06%	–	–	–
Small business	12,837	1.06%	–	–	–
Vacation	7,954	0.66%	–	–	–
Wedding	1,319	0.11%	–	–	–

Notes: The table uses the cleaned estimation sample of 1,213,853 loans originated in 2012–2018. Monetary amounts are in thousands of US dollars. FICO is the lower bound of the reported score range. Employment length is coded as zero for less than one year and ten for ten or more years.

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